NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Computing and Informatics | Department of Cyber Security

SIEM PROOF OF CONCEPT REPORT

Namibia E-Commerce Company Ltd Pty (NEC)

Applied Ethical Hacking AEH811S — Individual Assignment [100 Marks]

1. Executive Summary

This report documents the design and implementation of a Security Information and Event Management (SIEM) Proof of Concept (POC) for Namibia E-Commerce Company Ltd Pty (NEC). The objective was to demonstrate how a SIEM solution can be integrated into NEC's simulated network environment to improve cybersecurity posture, detect malicious activities in real time, and facilitate incident response.

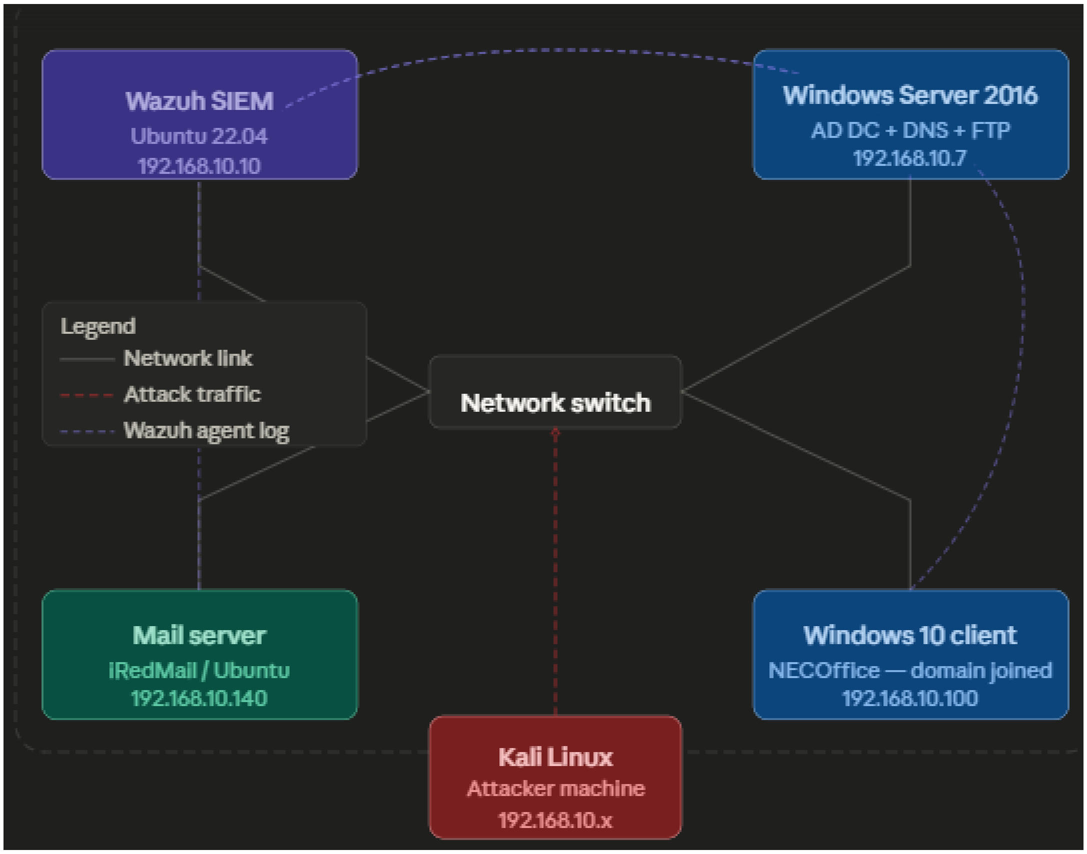
The POC was implemented using Wazuh as the SIEM platform, deployed across a virtualized lab environment on VMware Workstation. The environment includes a Windows Server 2016 Domain Controller with DNS and FTP services, an Ubuntu-based mail server running iRedMail with Roundcube webmail, a Windows 10 client machine joined to the domain, a Linux Ubuntu client, and a Kali Linux attacker machine used to simulate threats.

All log sources were integrated into Wazuh, enabling centralised monitoring, real-time alerting, and comprehensive incident response demonstrations.

2. NEC Network Emulation

2.1 Network Topology

All virtual machines are connected on the 192.168.10.0/24 subnet using VMware Workstation. The topology below illustrates the NEC simulated environment:



Network Legend:

- Dark Blue — Wazuh SIEM Server
- Blue — Windows Server 2016 (AD DC + DNS + FTP)
- Green — iRedMail Email Server
- Red — Attacker Machine (Kali Linux)
- Light Blue — Windows 10 Client

2.2 Virtual Machine Inventory

VM / Hostname	Role	OS	IP Address	Key Services
DC01	Domain Controller	Windows Server 2016	192.168.10.7	AD DS, DNS, FTP (IIS)
mail.nec.local	Mail Server	Ubuntu 22.04 LTS	192.168.10.140	iRedMail, Postfix, Dovecot,

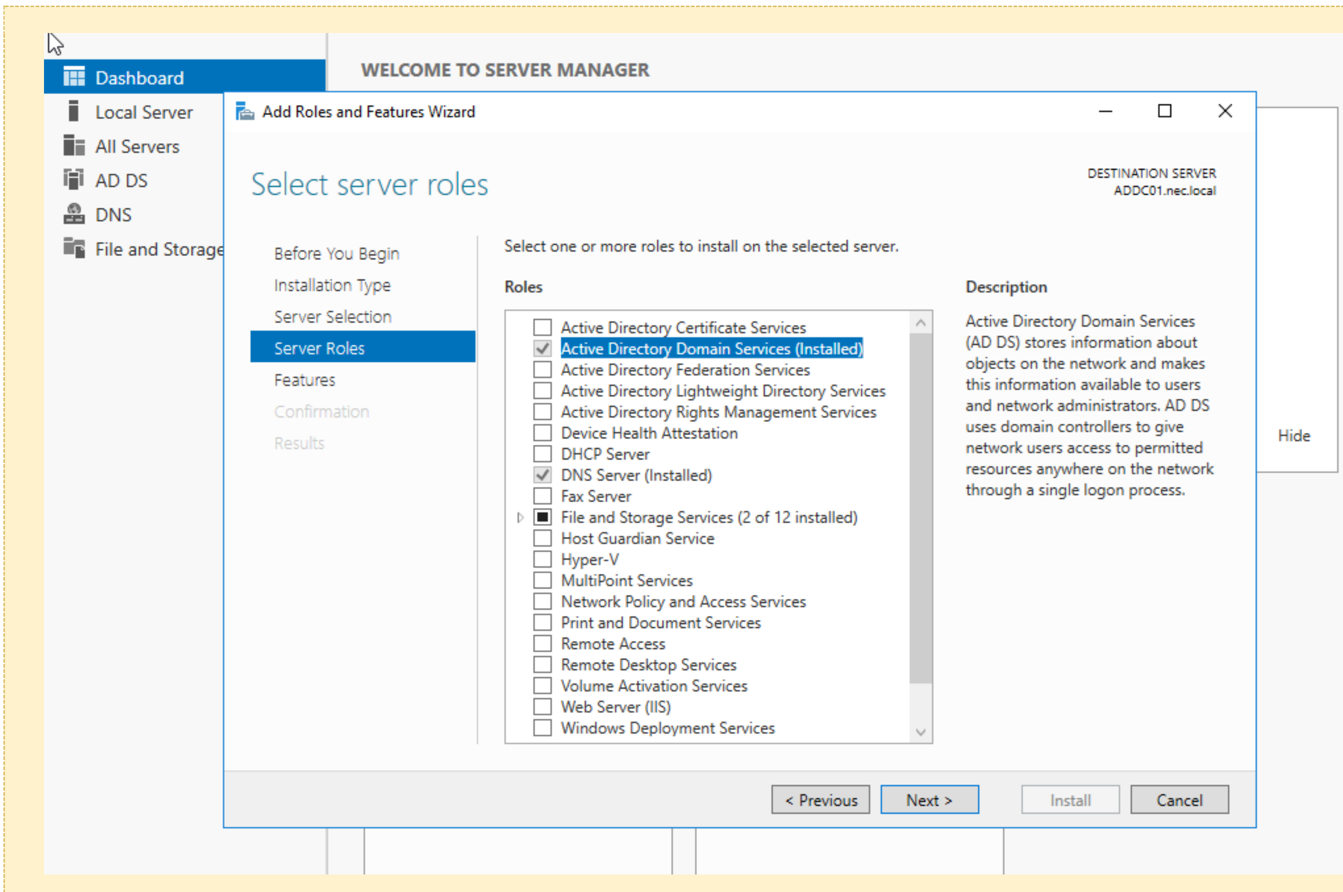
VM / Hostname	Role	OS	IP Address	Key Services
				Roundcube, Nginx
wazuh.nec.local	SIEM Server	Ubuntu 22.04 LTS	192.168.10.10	Wazuh Manager, Indexer, Dashboard
WIN10-CLIENT	Windows Client	Windows 10	192.168.10.100	Domain joined, Wazuh Agent
kali-attacker	Attacker Machine	Kali Linux	192.168.10.x	Hydra, Metasploit, Nmap

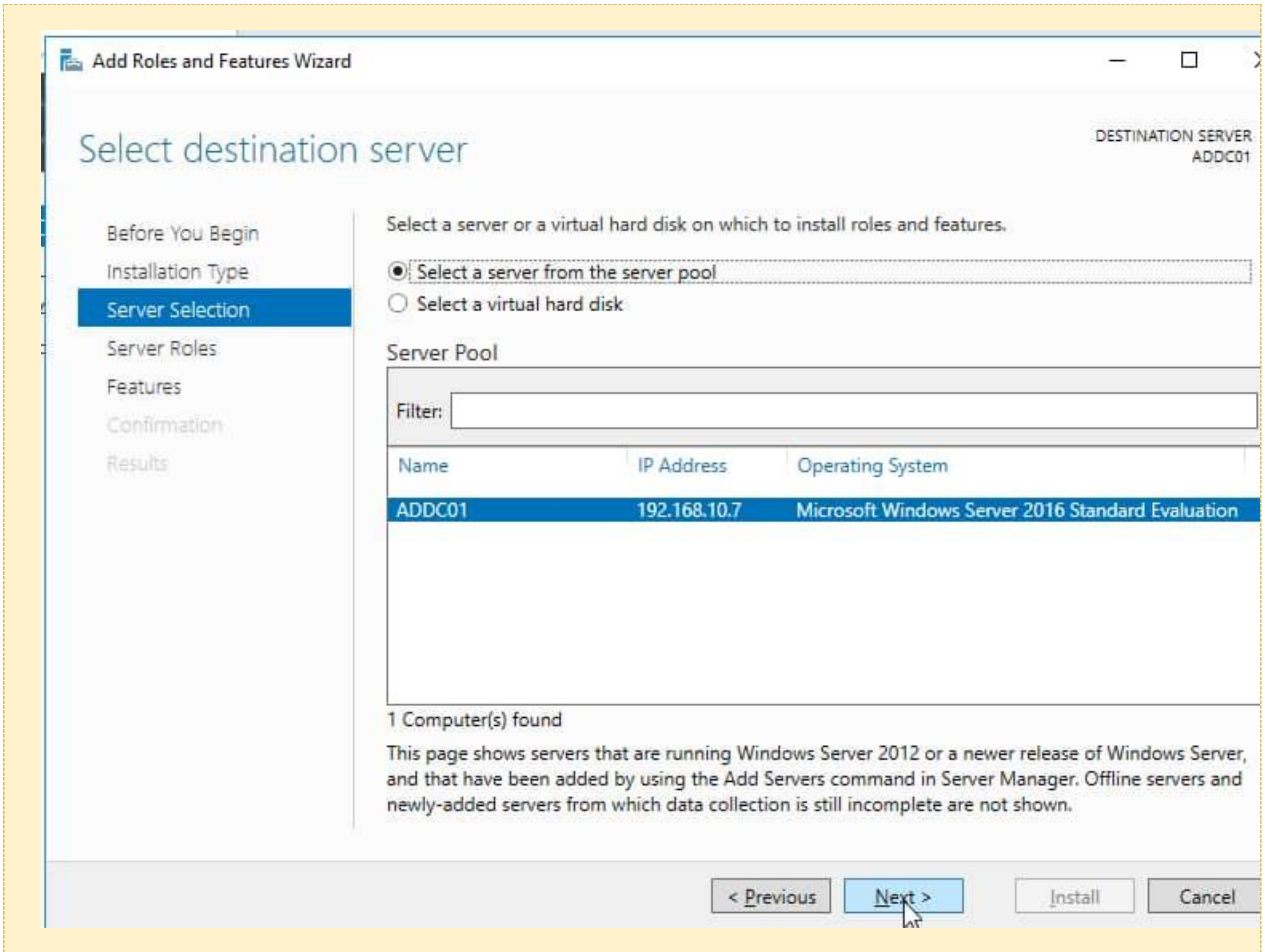
2.3 Active Directory Domain Setup

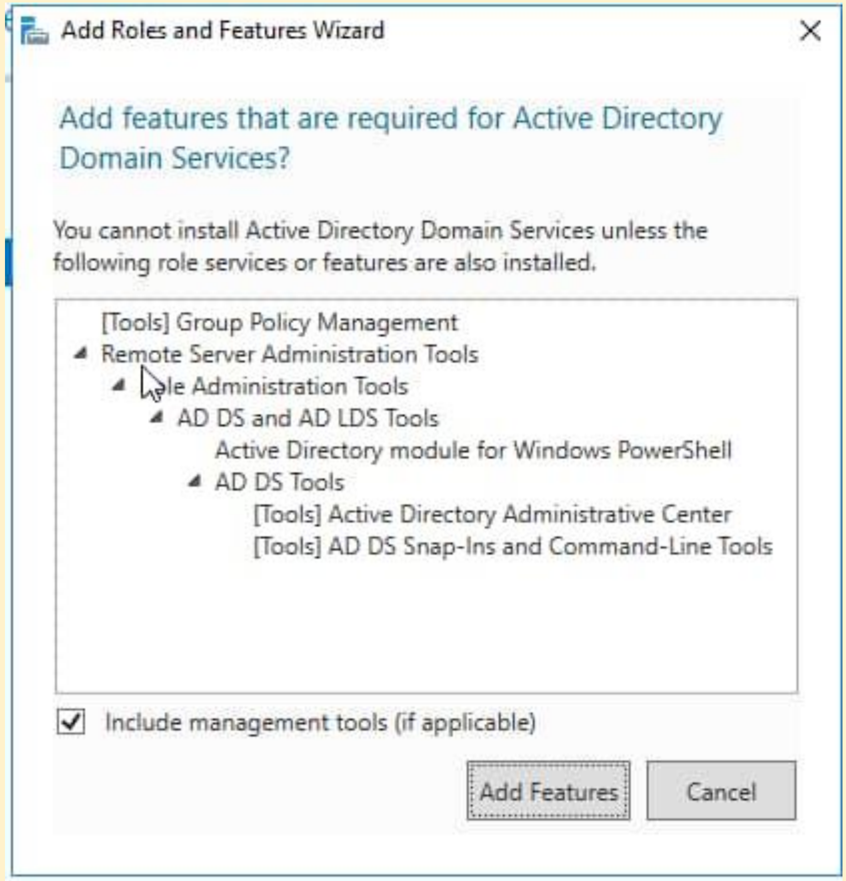
Windows Server 2016 was promoted to a Domain Controller for the nec.local domain. The following steps were performed:

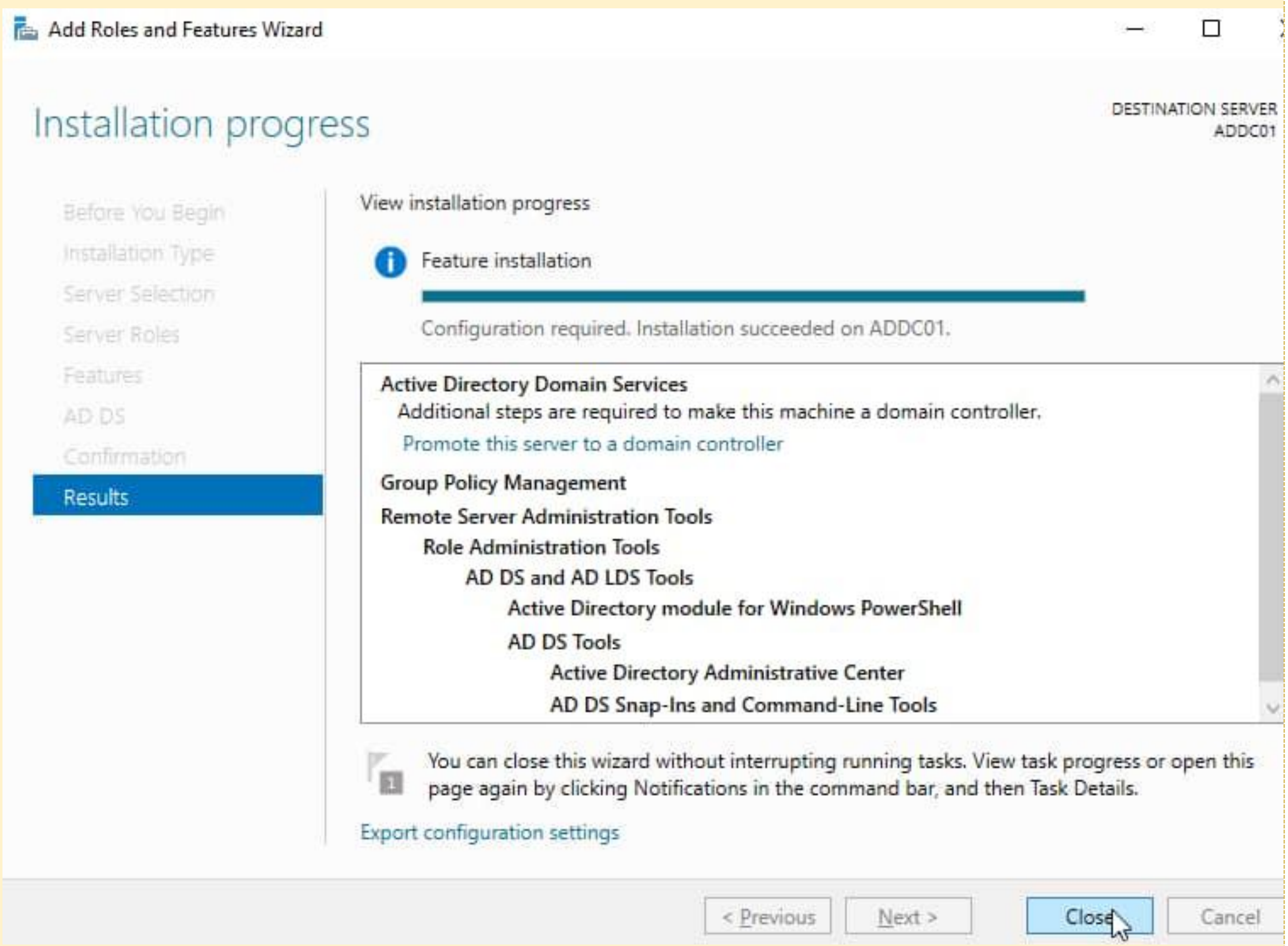
Step 1 — Installing AD DS Role

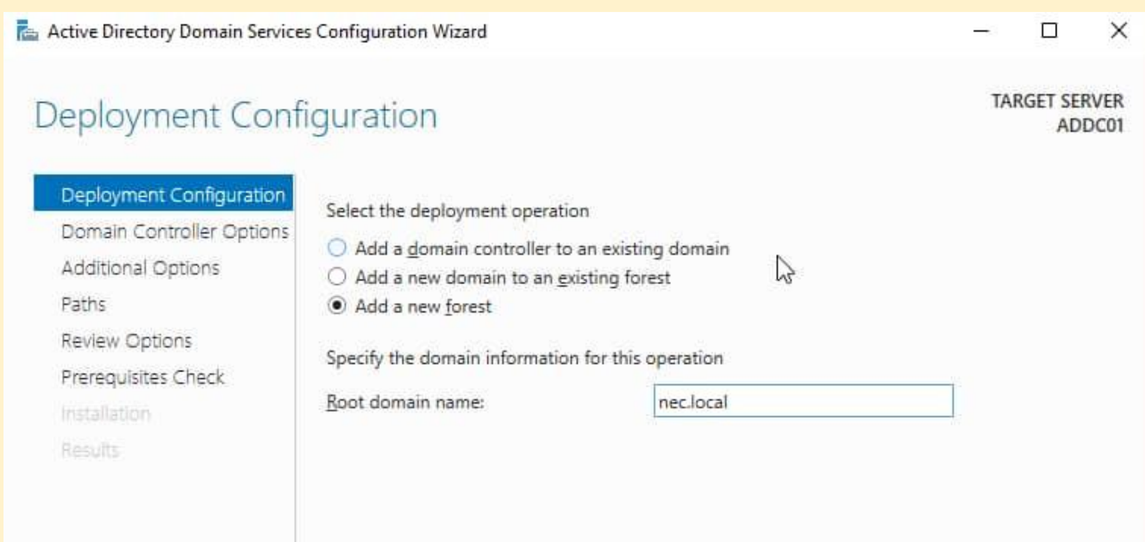
- Opened Server Manager on Windows Server 2016 (192.168.10.7)
- Navigated to Manage → Add Roles and Features
- Selected Active Directory Domain Services
- Completed installation and promoted server to Domain Controller
- Set root domain name to nec.local
- Set DSRM password and completed wizard

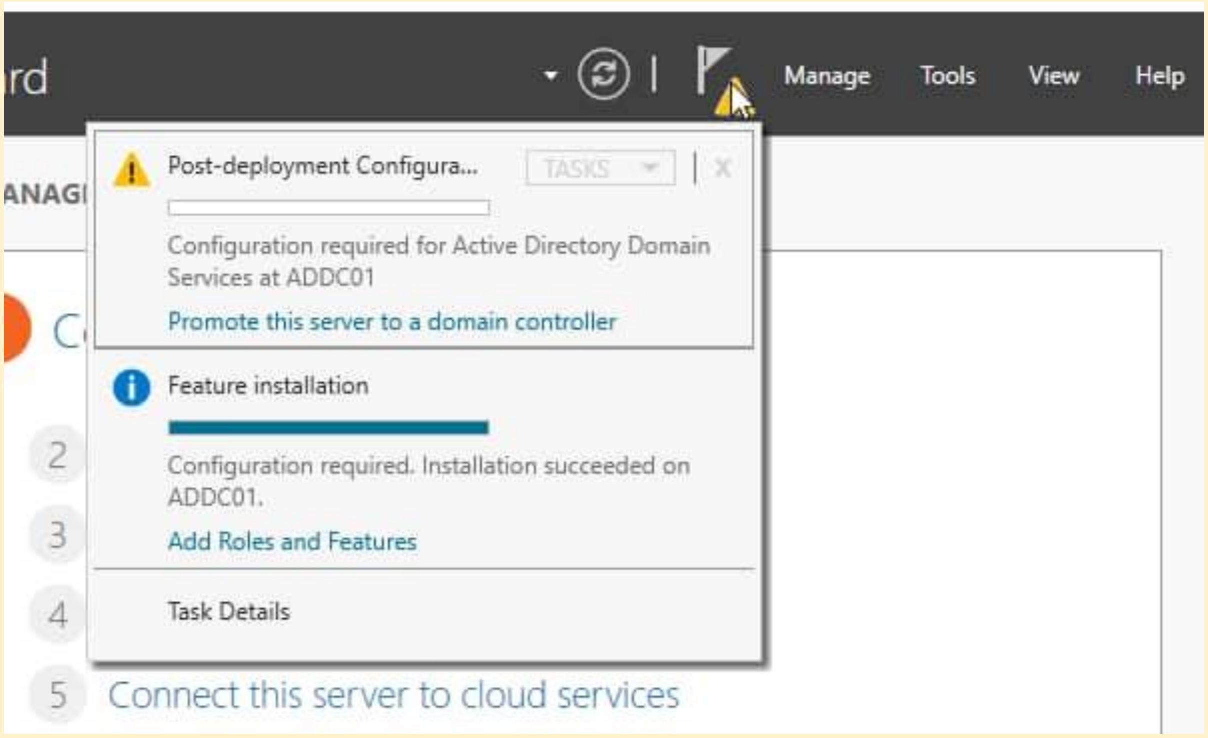



Add Roles and Features Wizard
Select destination server
DESTINATION SERVER
ADDC01
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select a server or a virtual hard disk on which to install roles and features.
Select a server from the server pool
Select a virtual hard disk
Server Pool
Filter:
Name
IP Address
Operating System
ADDC01
192.168.10.7
Microsoft Windows Server 2016 Standard Evaluation
1 Computer(s) found
This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.
< Previous
Next >
Install
Cancel


Add Roles and Features Wizard
Add features that are required for Active Directory Domain Services?
You cannot install Active Directory Domain Services unless the following role services or features are also installed.
[Tools] Group Policy Management
Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
Active Directory module for Windows PowerShell
AD DS Tools
[Tools] Active Directory Administrative Center
[Tools] AD DS Snap-Ins and Command-Line Tools
Include management tools (if applicable)
Add Features
Cancel


Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
ADDC01
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
Confirmation
Results
View installation progress
Feature installation
Configuration required. Installation succeeded on ADDC01.
Active Directory Domain Services
Additional steps are required to make this machine a domain controller.
Promote this server to a domain controller
Group Policy Management
Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
Active Directory module for Windows PowerShell
AD DS Tools
Active Directory Administrative Center
AD DS Snap-Ins and Command-Line Tools
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel


Active Directory Domain Services Configuration Wizard
Deployment Configuration
TARGET SERVER
ADDC01
Deployment Configuration
Domain Controller Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Select the deployment operation
Add a domain controller to an existing domain
Add a new domain to an existing forest
Add a new forest
Specify the domain information for this operation
Root domain name:
nec.local

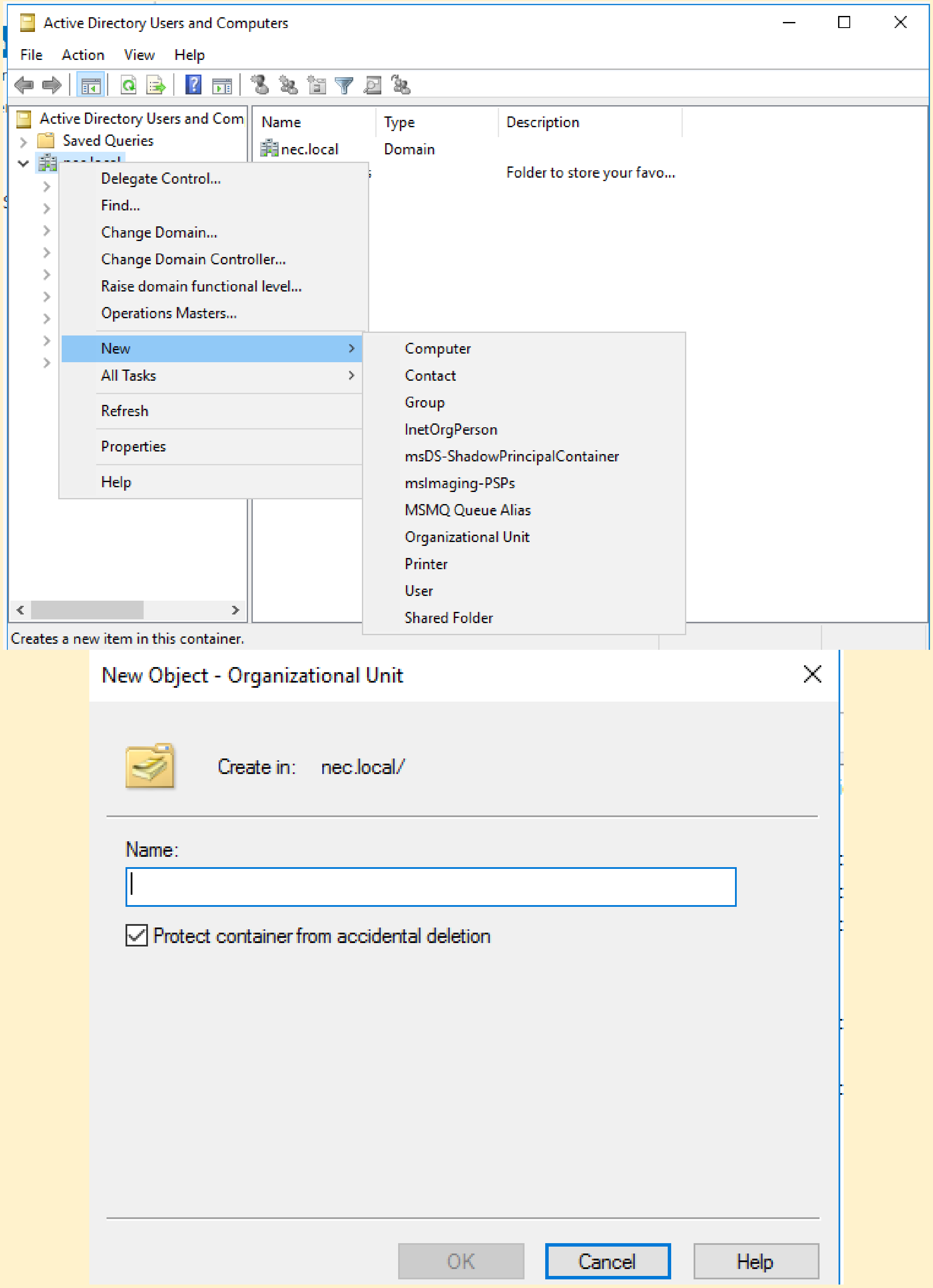


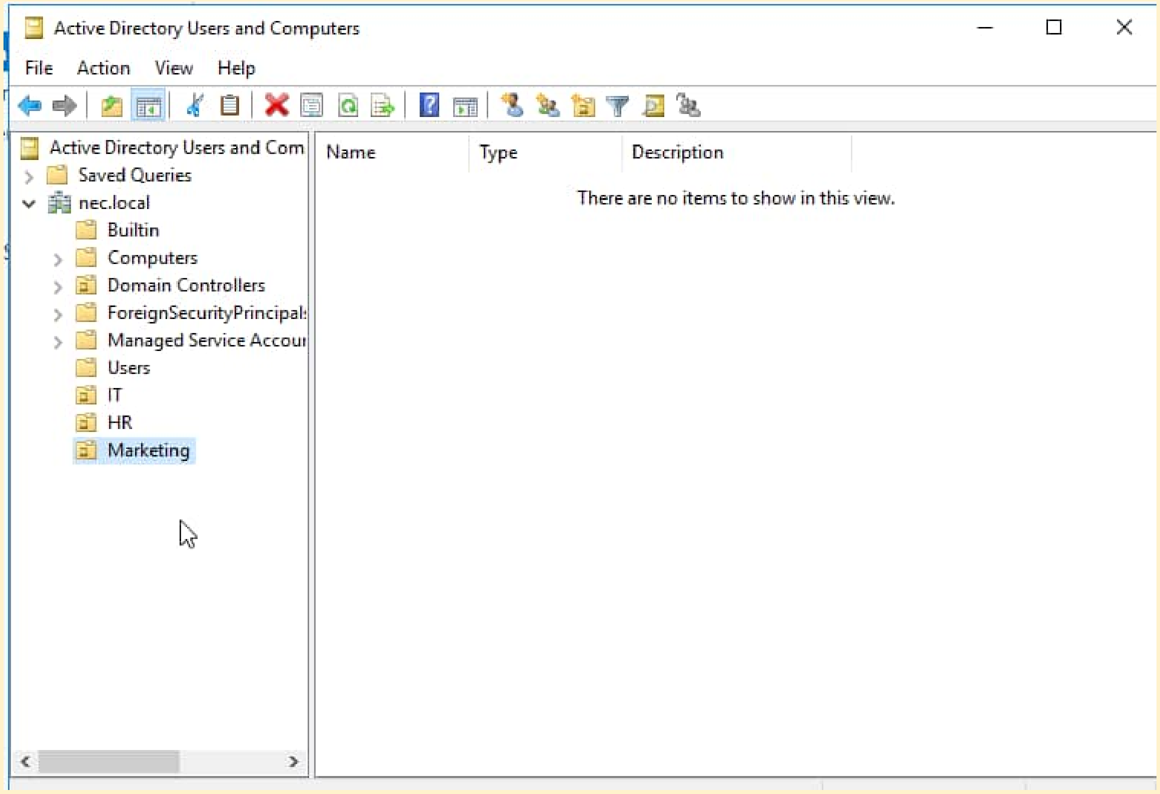
Screenshot: Active Directory Domain Services Configuration Wizard — Forest: nec.local

Step 2 — Creating Organisational Units (OUs)

Three OUs were created in Active Directory Users and Computers to represent NEC's departments:

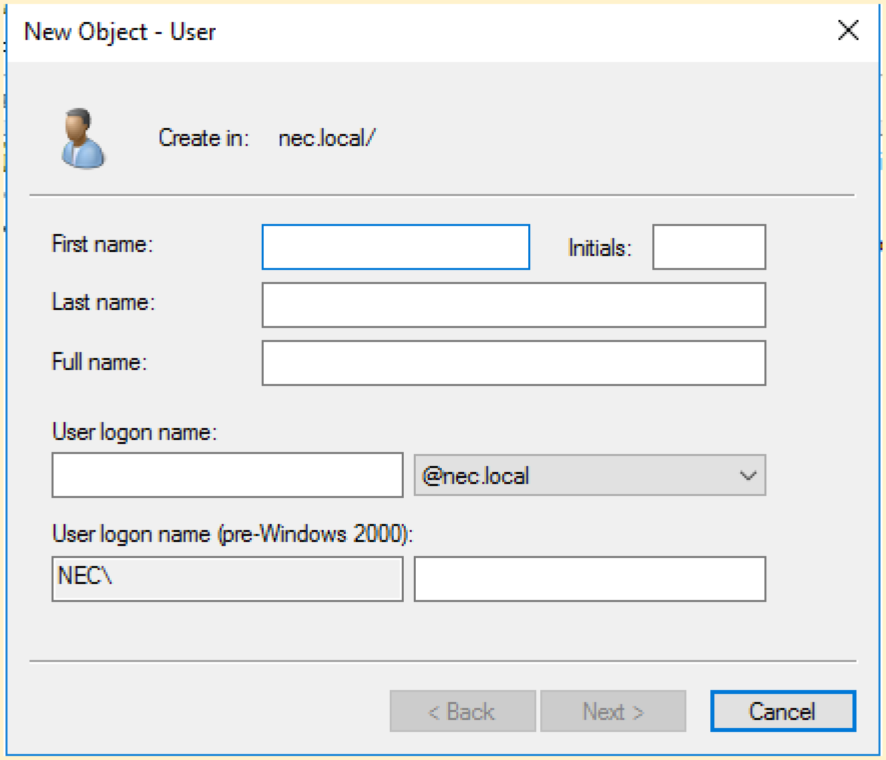
Organisational Unit	Purpose
IT	Information Technology department users
HR	Human Resources department users
Marketing	Marketing department users

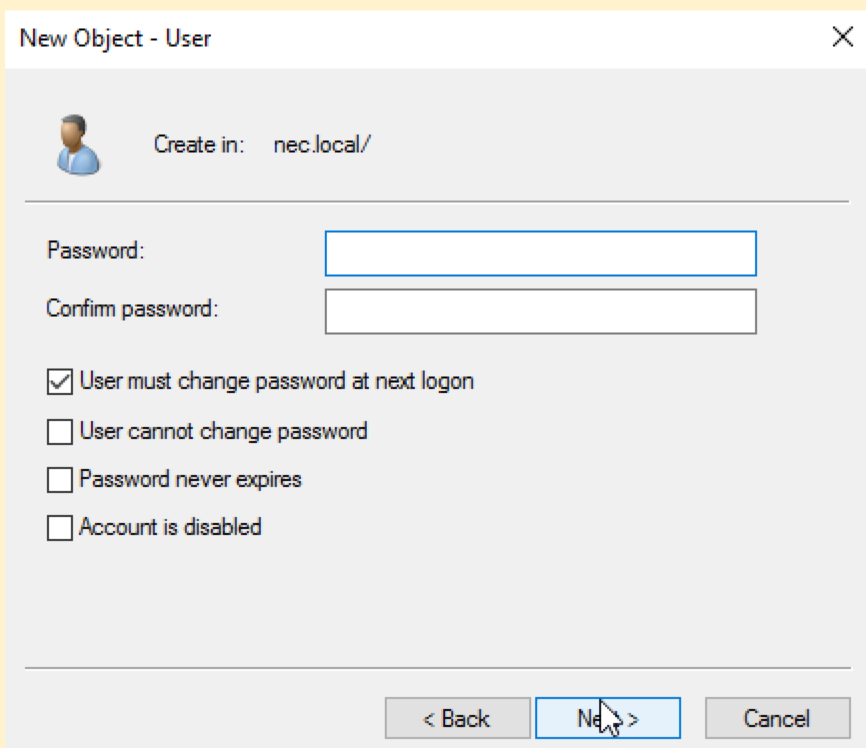

Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
Name
Type
Description
nec.local
Domain
Folder to store your favo...
Delegate Control...
Find...
Change Domain...
Change Domain Controller...
Raise domain functional level...
Operations Masters...
New
All Tasks
Refresh
Properties
Help
Computer
Contact
Group
InetOrgPerson
msDS-ShadowPrincipalContainer
msImaging-PSPs
MSMQ Queue Alias
Organizational Unit
Printer
User
Shared Folder
Creates a new item in this container.
New Object - Organizational Unit
Create in: nec.local/
Name:
Protect container from accidental deletion
OK
Cancel
Help



Step 3 — Creating Domain User Accounts

Full Name	Username	Email	OU	Password
John Smith	john.smith	john.smith@nec.local	IT	User@NEC2026!
Sarah Jones	sarah.jones	sarah.jones@nec.local	HR	User@NEC2026!
David Brown	david.brown	david.brown@nec.local	Marketing	User@NEC2026!


New Object - User
Create in: nec.local/
First name:
Initials:
Last name:
Full name:
User logon name:
@nec.local
User logon name (pre-Windows 2000):
NEC\
< Back
Next >
Cancel


New Object - User
Create in: nec.local/
Password:
Confirm password:
User must change password at next logon
User cannot change password
Password never expires
Account is disabled
< Back
Cancel

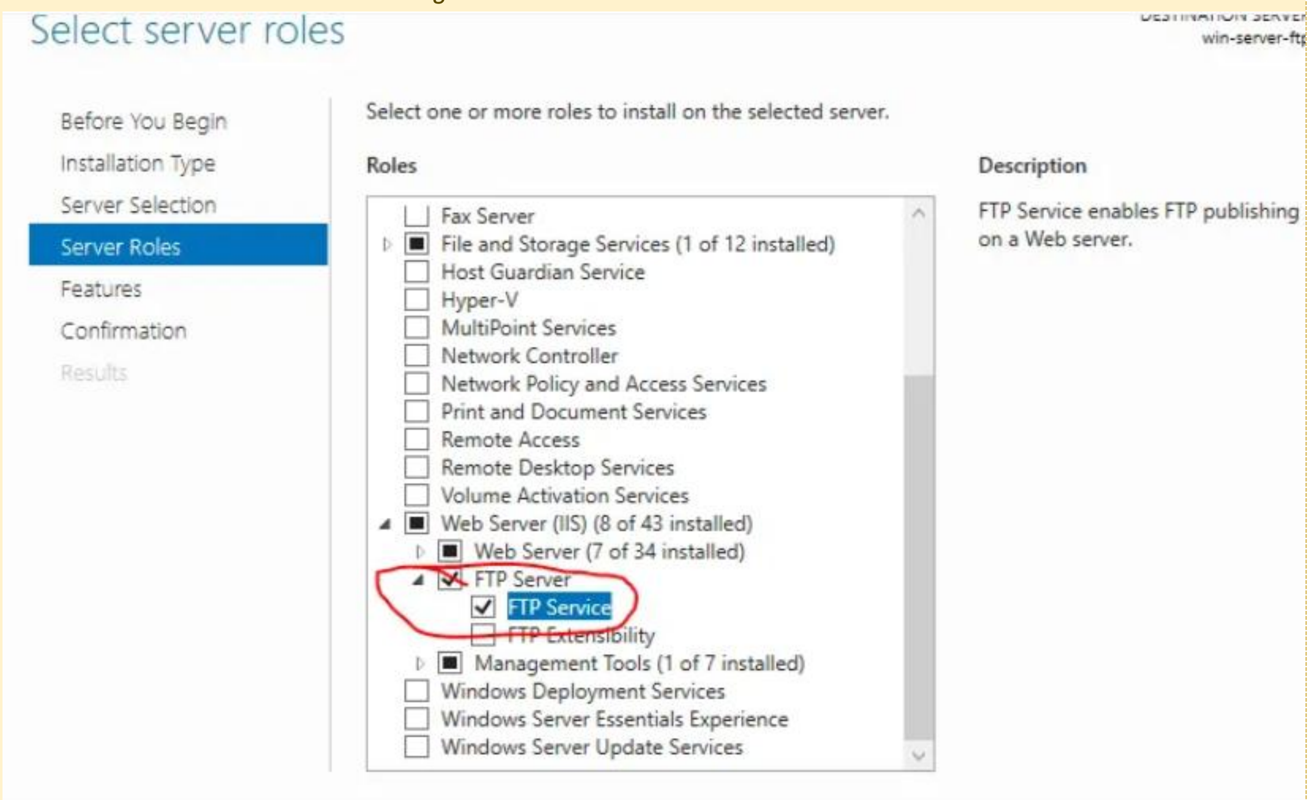
2.5 FTP Server Configuration

IIS FTP Server was installed on Windows Server 2016 to provide file sharing services for NEC departments.

Step 1 — Installing IIS and FTP Role

- Server Manager → Add Roles and Features → Web Server (IIS)
- Expanded and selected: FTP Server, FTP Service, FTP Extensibility
- Completed installation

Screenshot: Server Manager — Add Roles and Features — FTP Server role selected

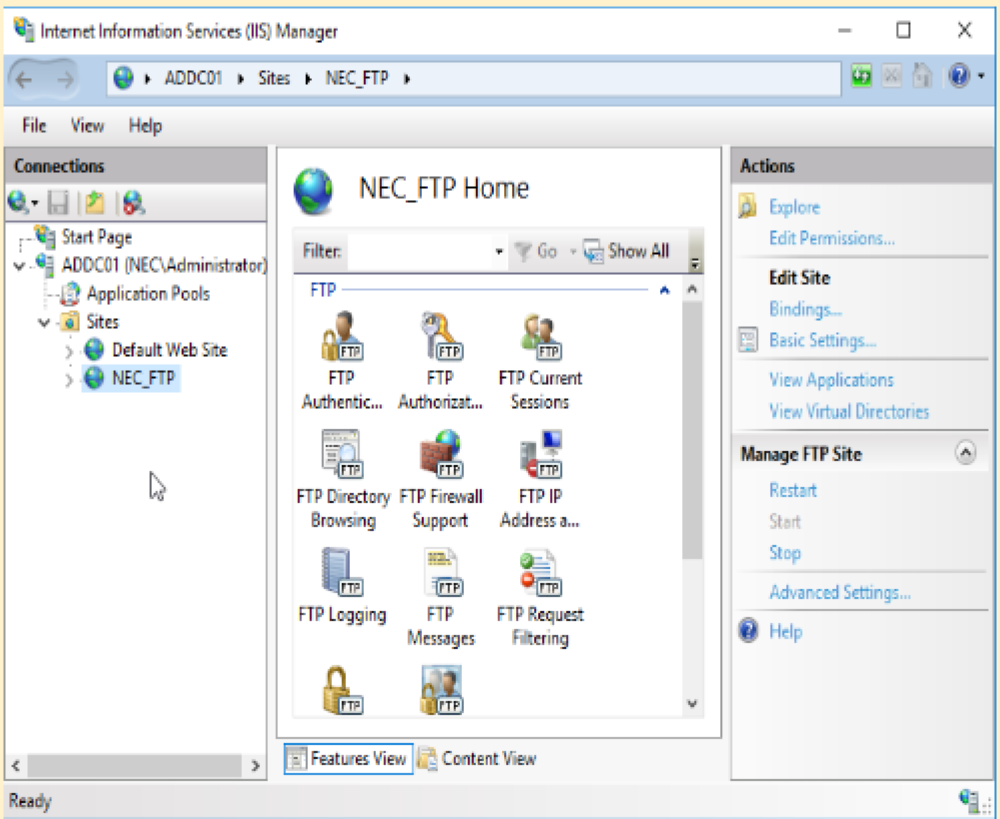


Step 2 — Creating FTP Directories

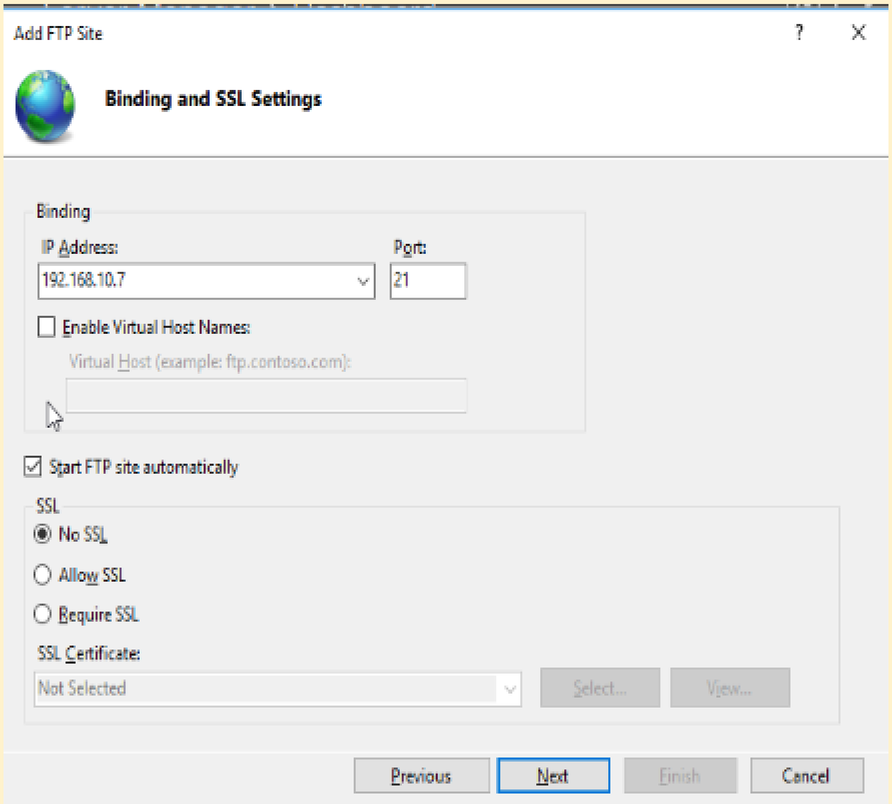
Directory	Path	Assigned Department
NEC FTP Root	C:\FTP	All users
IT Folder	C:\FTP\IT	John Smith (IT)
HR Folder	C:\FTP\HR	Sarah Jones (HR)
Marketing Folder	C:\FTP\Marketing	David Brown (Marketing)

Step 3 — FTP Site Configuration

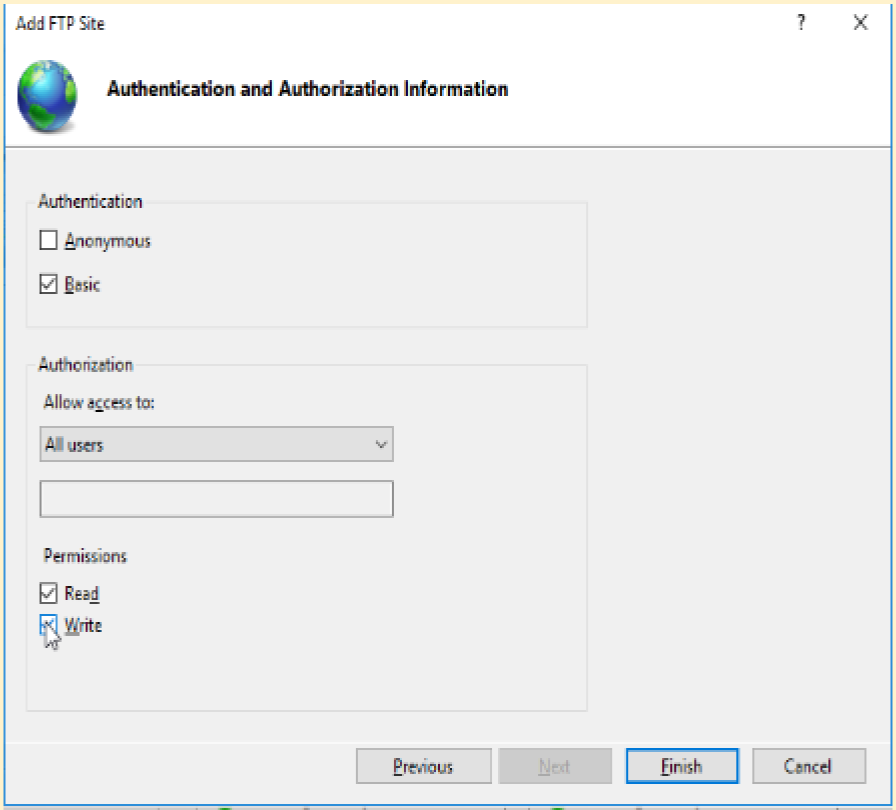
Setting	Value
FTP Site Name	NEC FTP
Physical Path	C:\FTP
IP Address	192.168.10.7
Port	21
SSL	No SSL (Lab environment)
Authentication	Basic Authentication
Permissions	Read and Write



Screenshot: IIS Manager — NEC FTP site configured and running



Screenshot: FTP site Binding settings showing IP 192.168.10.7 on port 21



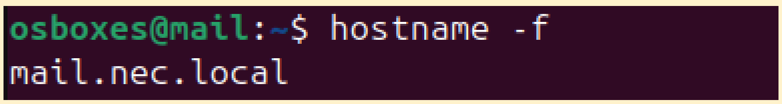
Screenshot: FTP Authentication and Authorization settings

2.6 Email Server Setup (iRedMail)

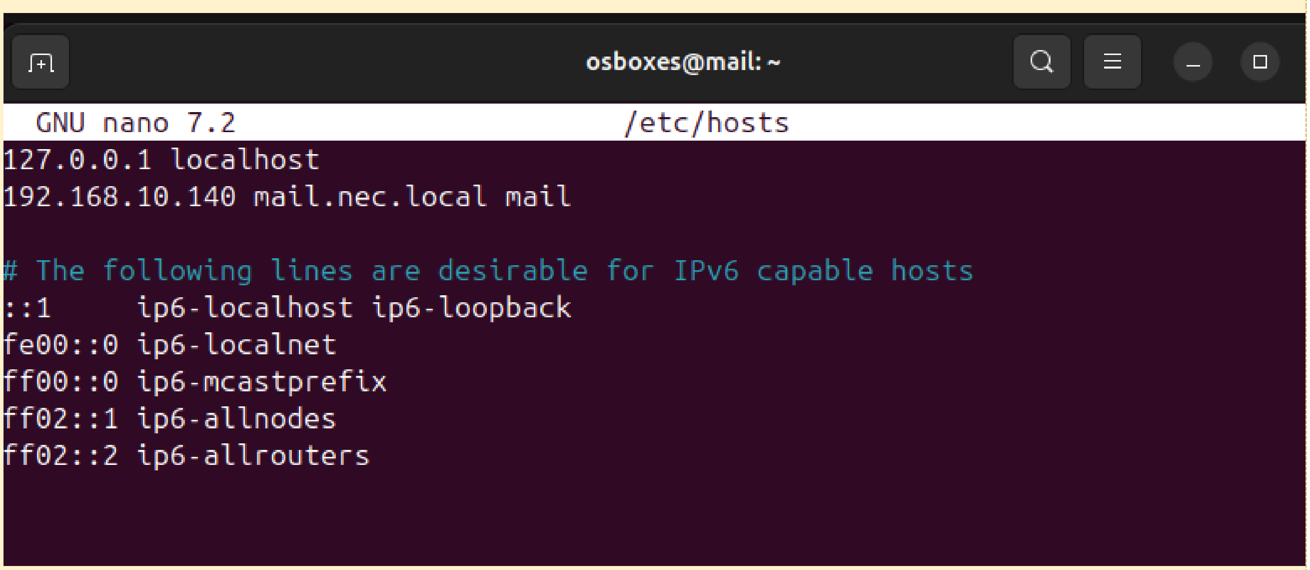
An Ubuntu 22.04 LTS server was configured as the corporate email server for NEC using iRedMail. iRedMail is an open-source mail server solution that provides a complete email stack.

Step 1 — Pre-Installation Setup

- Set static IP address to 192.168.10.140 using Netplan
- Set FQDN hostname: mail.nec.local
- Updated /etc/hosts: 192.168.10.140   mail.nec.local   mail
- Updated and upgraded Ubuntu: sudo apt update && sudo apt upgrade -y
- Disabled AppArmor for lab compatibility



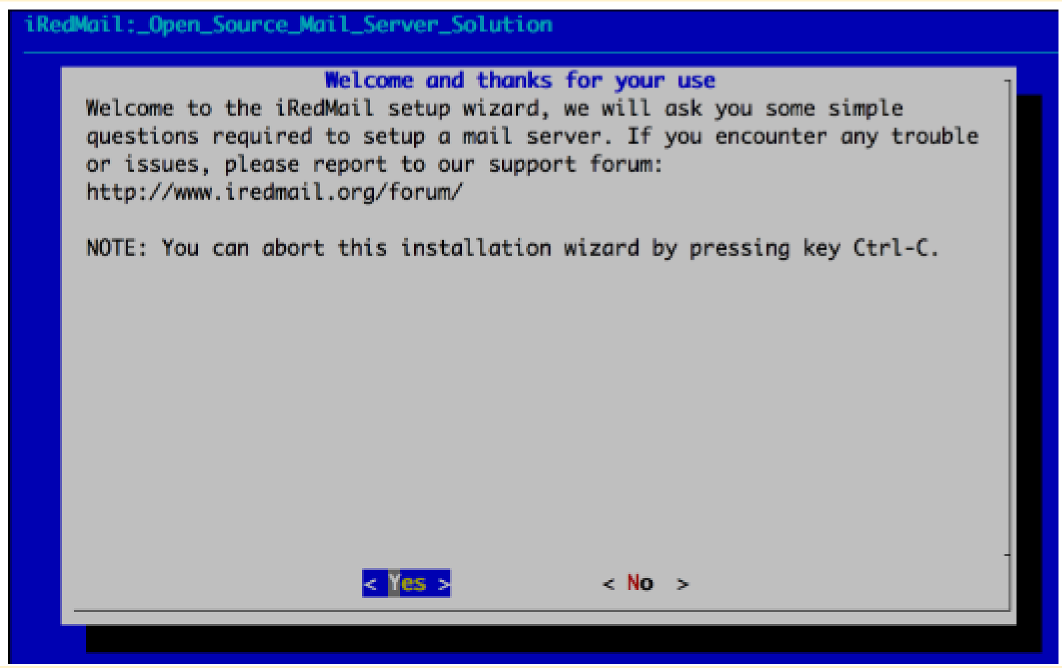
Screenshot: Terminal showing hostname -f returning mail.nec.local



Screenshot: /etc/hosts file showing correct IP to hostname mapping

Step 2 — Downloading and Installing iRedMail

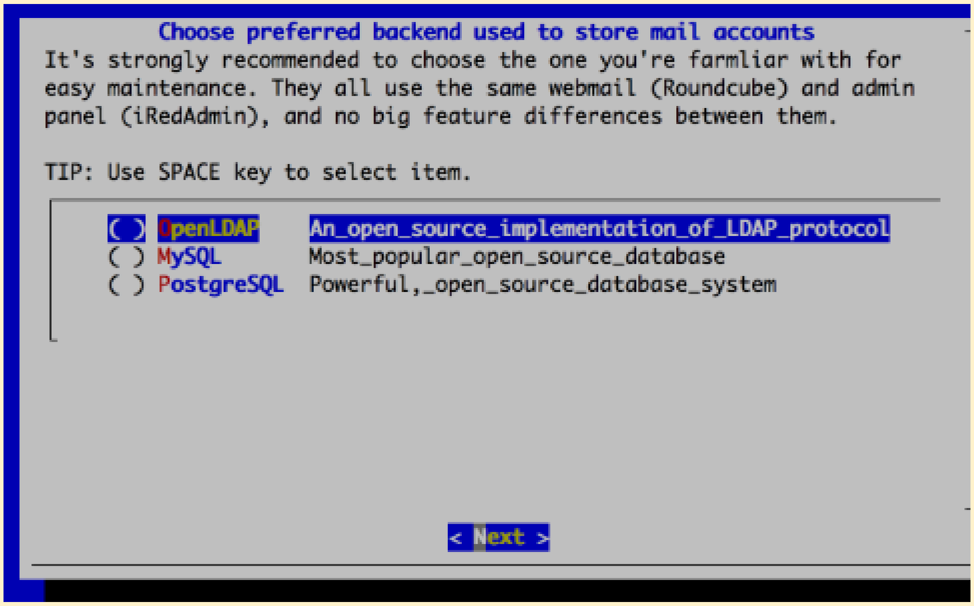
- Downloaded iRedMail 1.7.1 from GitHub
- Extracted archive and ran installer: sudo bash iRedMail.sh



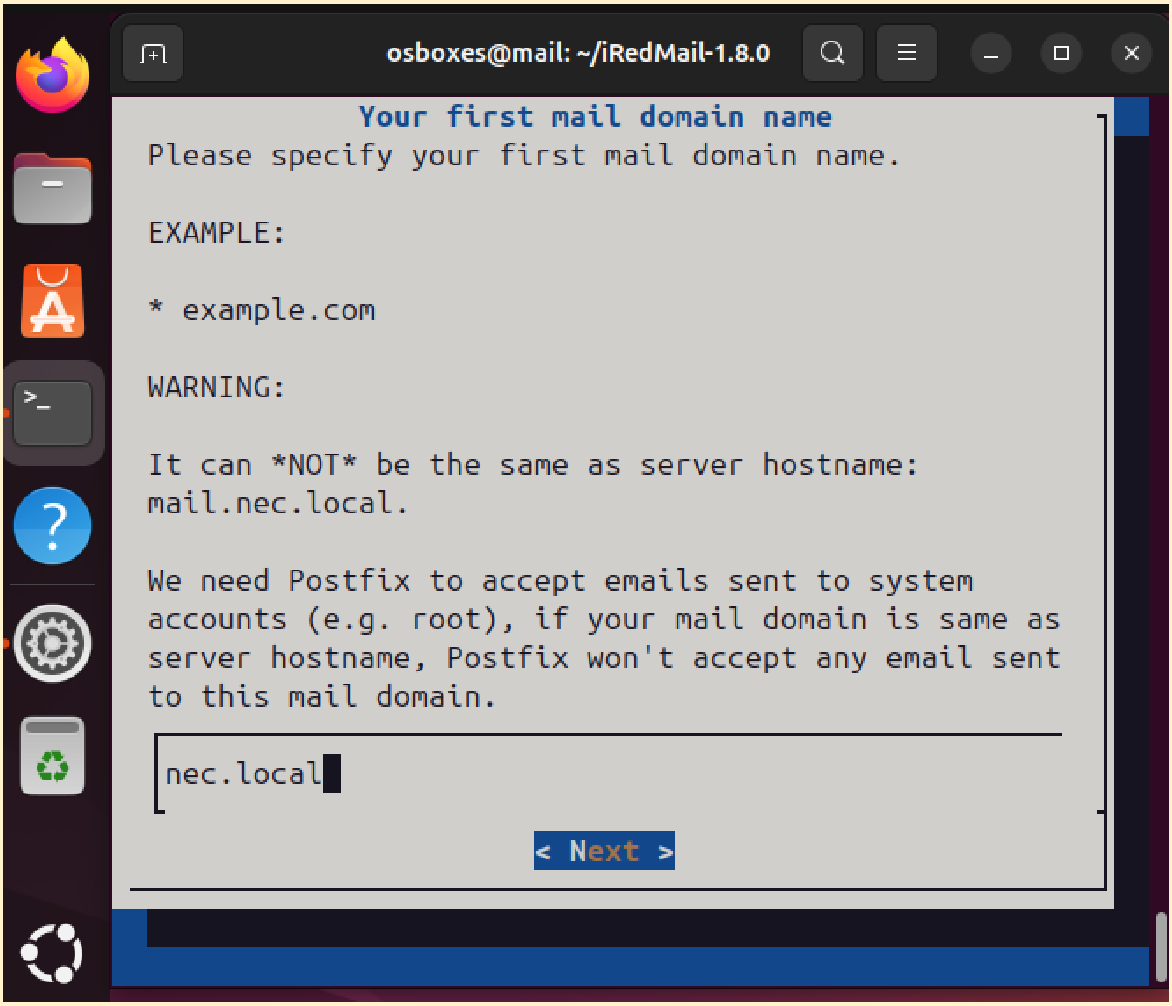
Screenshot: iRedMail installer wizard — Welcome screen

Step 3 — iRedMail Wizard Selections

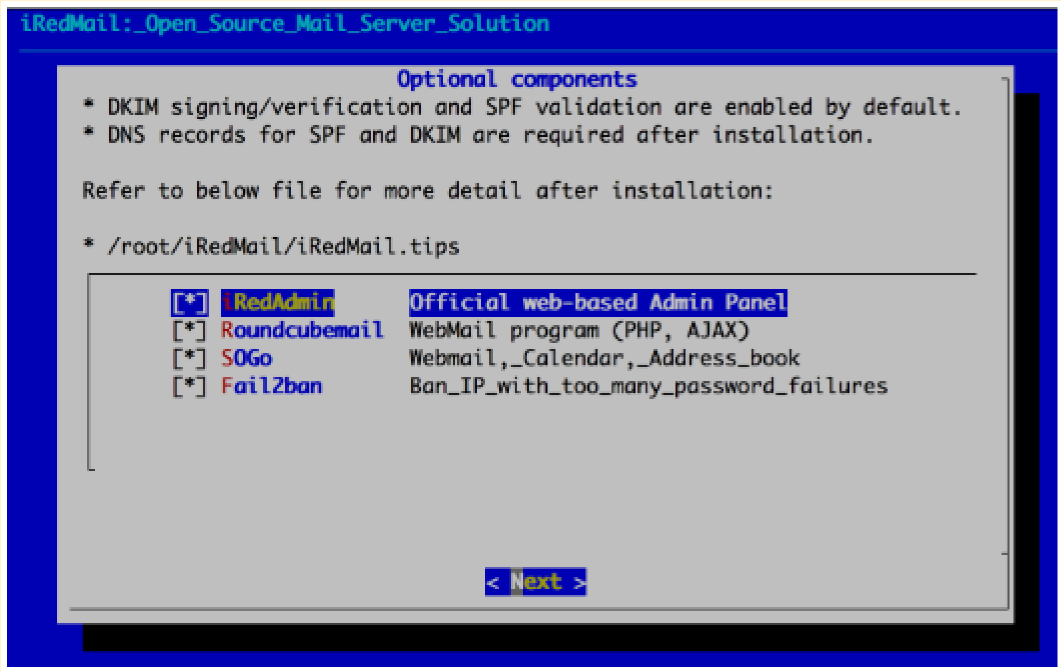
Wizard Prompt	Selection Made
Mail storage path	/var/vmail (default)
Web server	Nginx
Mail account backend	OpenLDAP
LDAP suffix	dc=nec,dc=local
Mail domain	nec.local
Admin password	Set during installation
Roundcubemail	Yes — selected
iRedAdmin	Yes — selected
Fail2Ban	Yes — selected
ClamAV	No — skipped to save RAM
SOGo Groupware	No — not required



Screenshot: iRedMail wizard — Backend selection (OpenLDAP)



Screenshot: iRedMail wizard — Mail domain set to nec.local

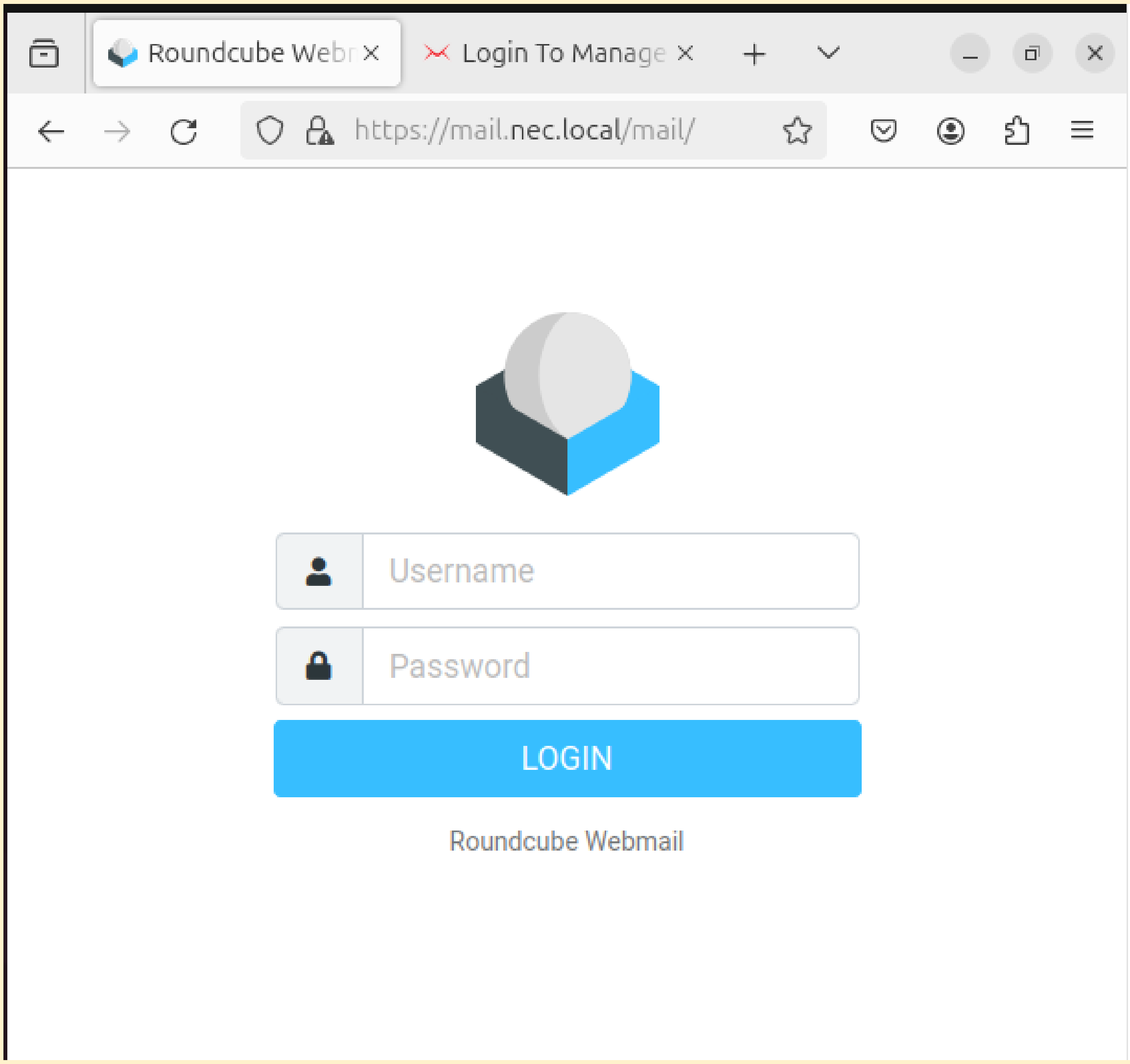


Screenshot: iRedMail wizard — Optional components selection screen

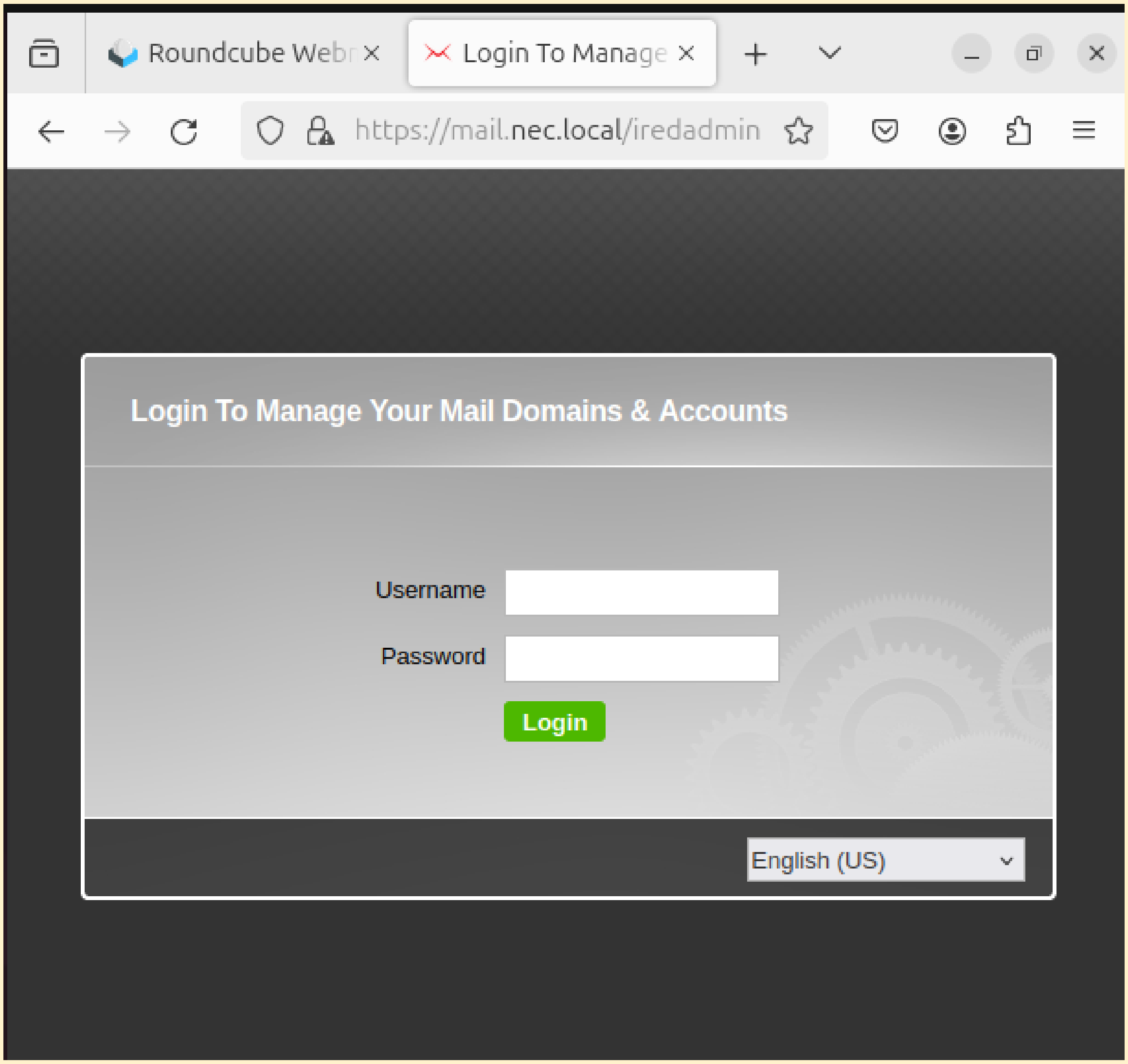
Screenshot: iRedMail installation completing successfully

Step 5 — Accessing iRedMail Web Interfaces

Interface	URL	Credentials
Roundcube Webmail	https://192.168.10.140/mail	User email + password
iRedAdmin Panel	https://192.168.10.140/iredadmin	postmaster@nec.local



Screenshot: Roundcube webmail login page at https://192.168.10.140/mail

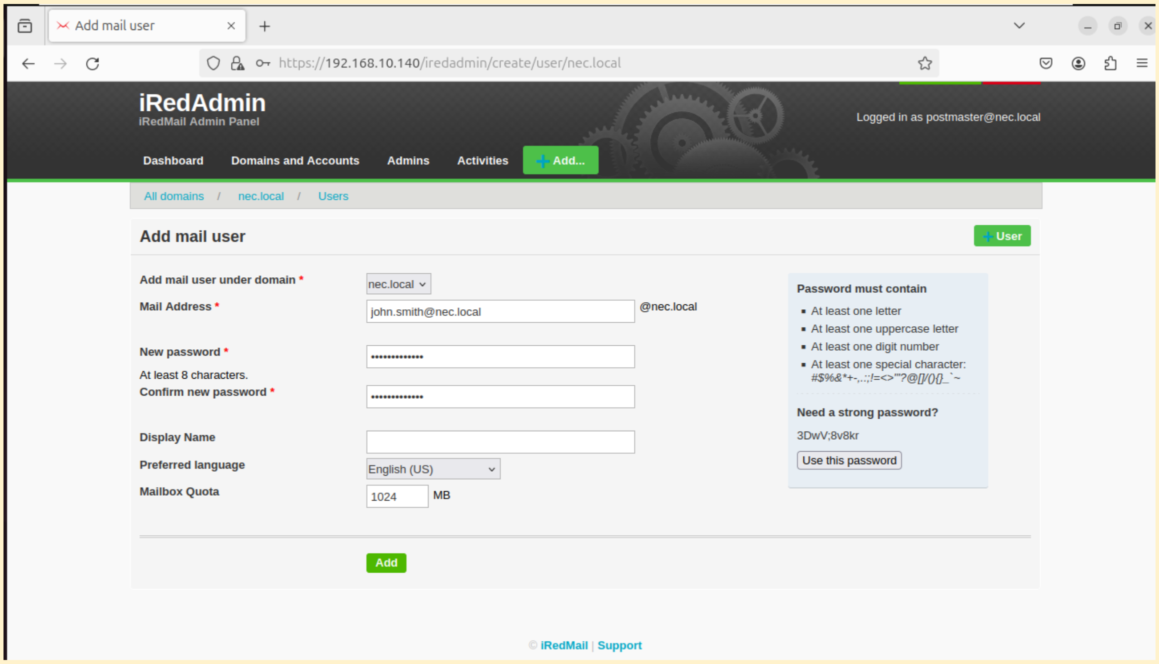


Screenshot: iRedAdmin login page at https://192.168.10.140/iredadmin

Step 6 — Creating Mail User Accounts in iRedAdmin

Three mail accounts were created in iRedAdmin to match the NEC department users:

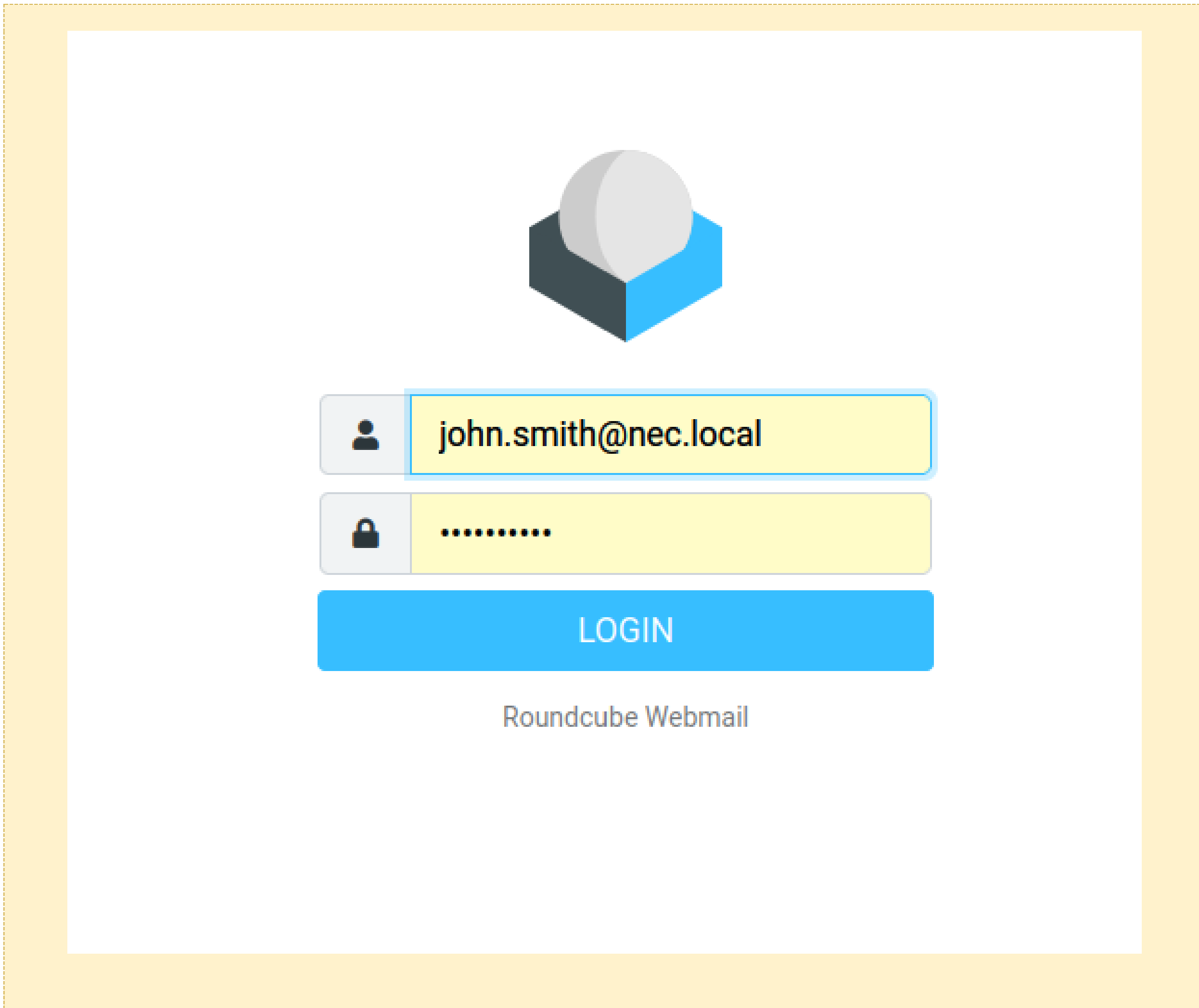
Full Name	Email Address	Department
John Smith	john.smith@nec.local	IT
Sarah Jones	sarah.jones@nec.local	HR
David Brown	david.brown@nec.local	Marketing

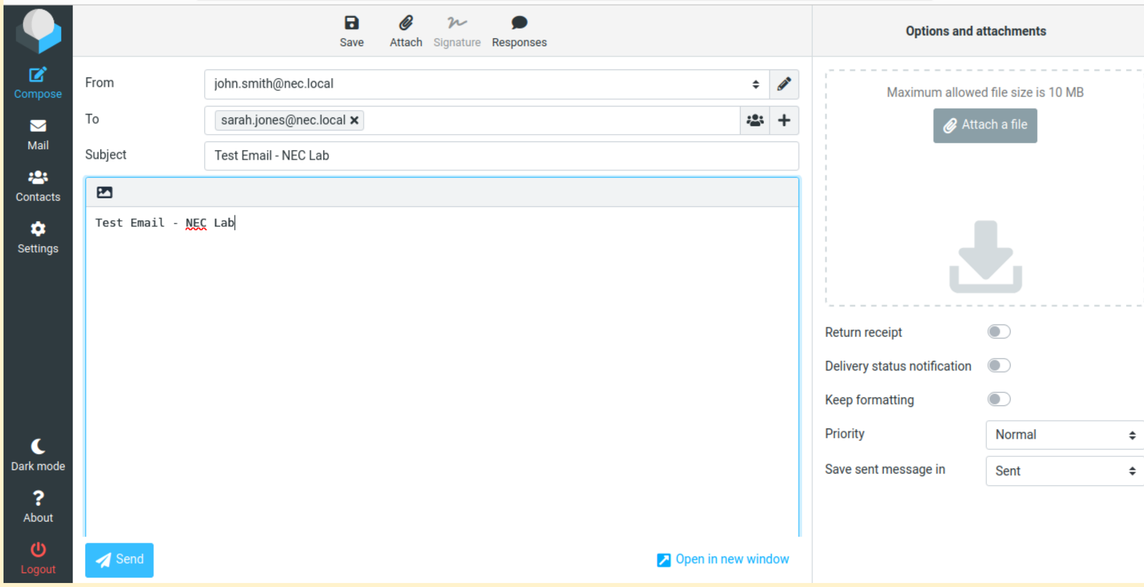


Screenshot: iRedAdmin — Add User form for john.smith@nec.local

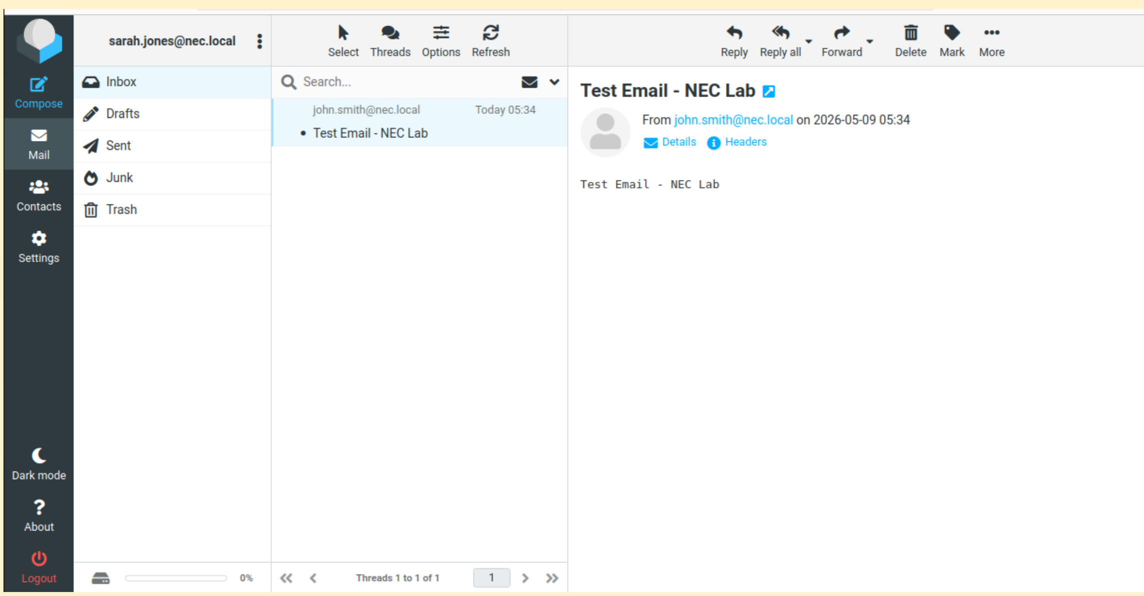
Step 7 — Testing Email Functionality

- Logged into Roundcube as john.smith@nec.local
- Composed and sent test email to sarah.jones@nec.local
- Logged out and logged in as sarah.jones@nec.local
- Confirmed email received in inbox


john.smith@nec.local
LOGIN
Roundcube Webmail



Screenshot: Roundcube — Composing email to sarah.jones@nec.local



Screenshot: Roundcube — Sarah Jones inbox showing received email from John Smith

2.7 Windows 10 Client — Domain Join

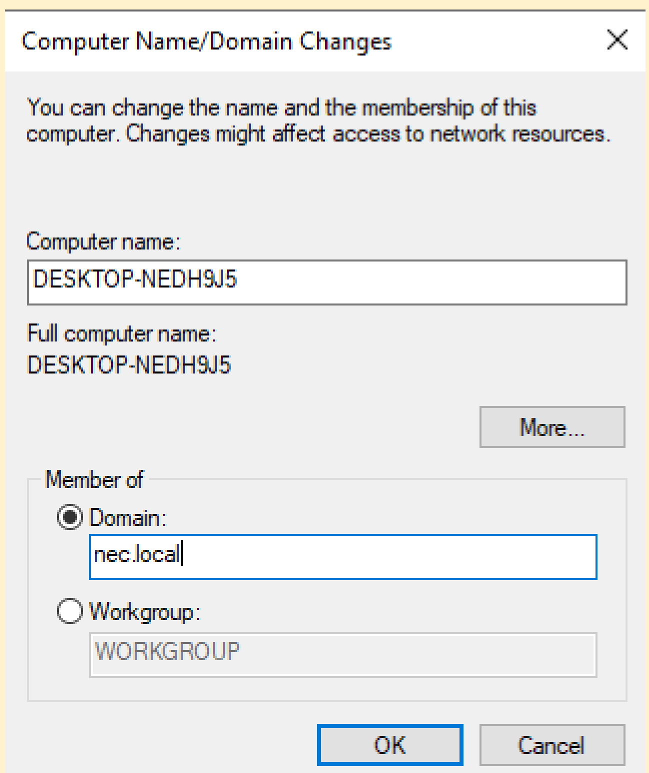
A Windows 10 client machine was configured and joined to the nec.local domain.

Step 1 — Network Configuration

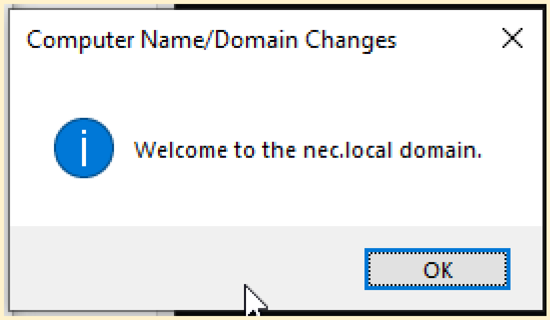
Setting	Value
IP Address	192.168.10.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
Preferred DNS	192.168.10.7 (Domain Controller)

Step 2 — Joining the Domain

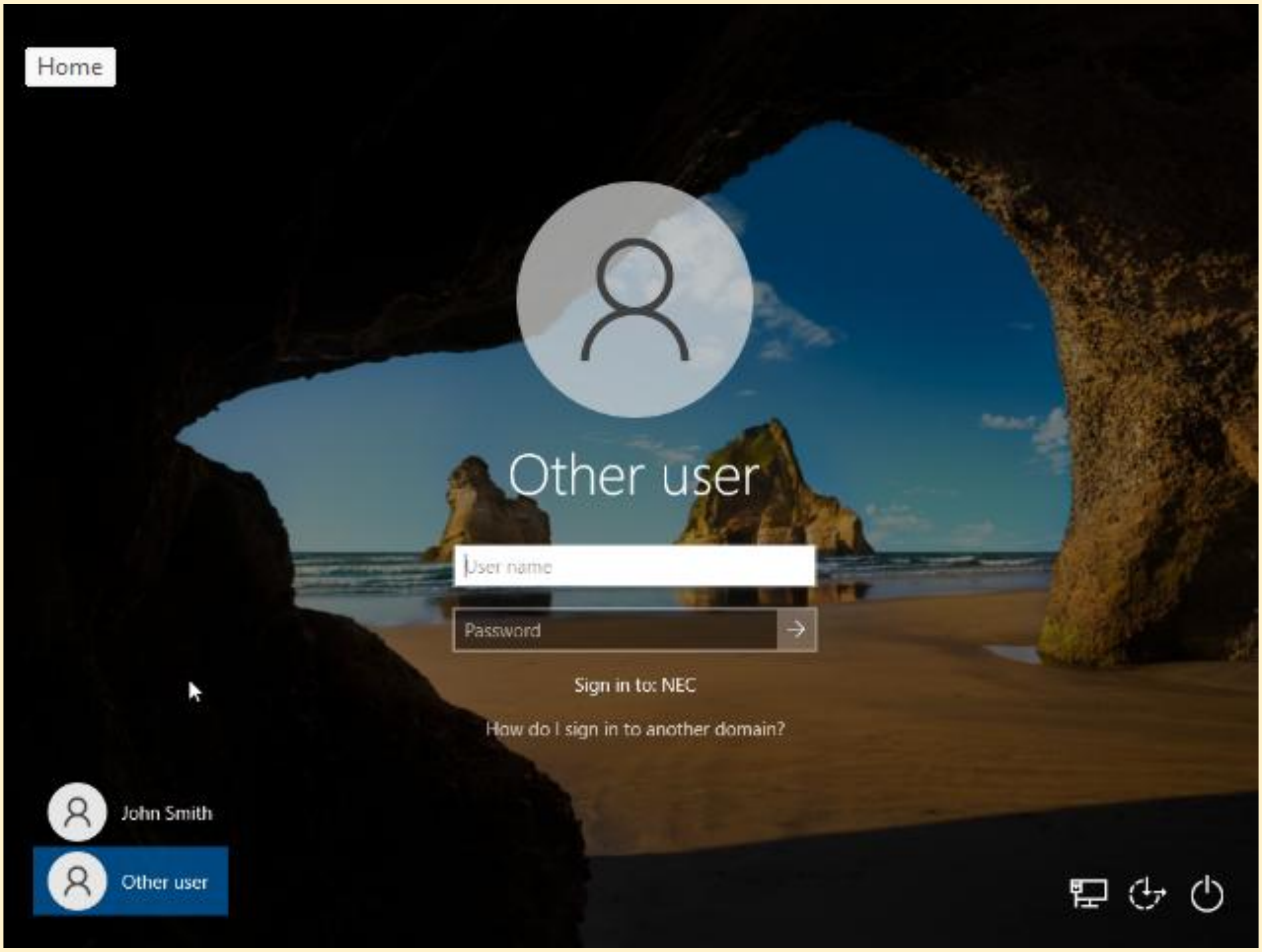
- Right-clicked This PC → Properties → Change Settings → Change
- Selected Domain and typed: nec.local
- Entered domain administrator credentials
- Restarted when prompted



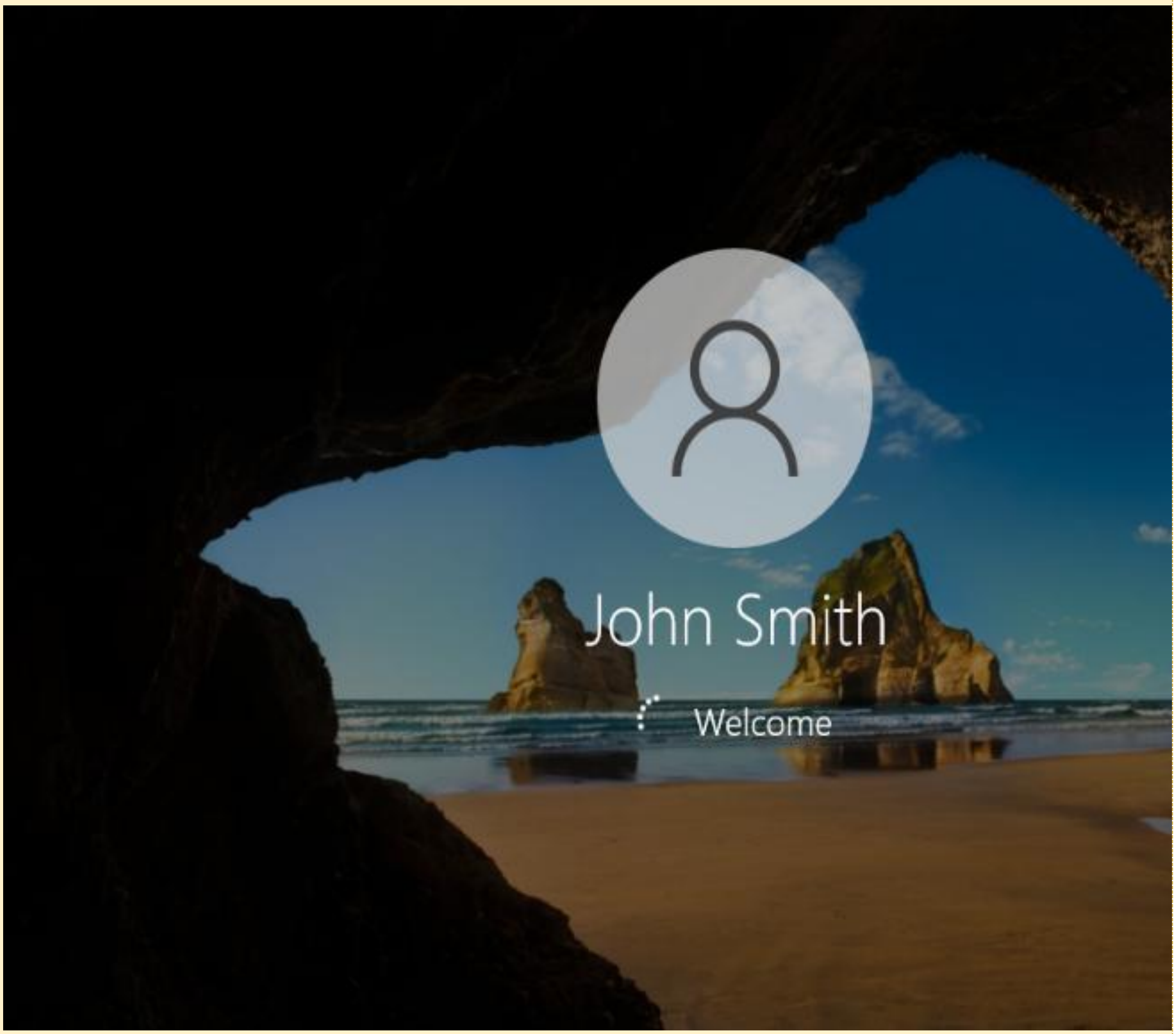
Screenshot: Windows 10 — System Properties showing domain join dialog with nec.local



Screenshot: Windows 10 — Welcome to nec.local domain message



Screenshot: Windows 10 — Login screen showing Other User option for domain login



Screenshot: Windows 10 — Logged in as john.smith@nec.local successfully

3. SIEM Integration and Configuration

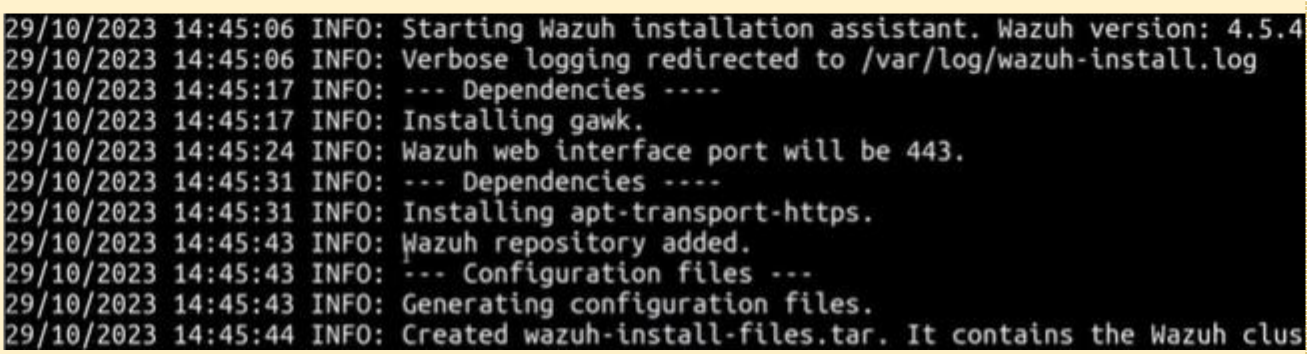
3.1 Wazuh Installation

Wazuh was installed on a dedicated Ubuntu 24.04 LTS server (192.168.10.10) using the official all-in-one installation script.

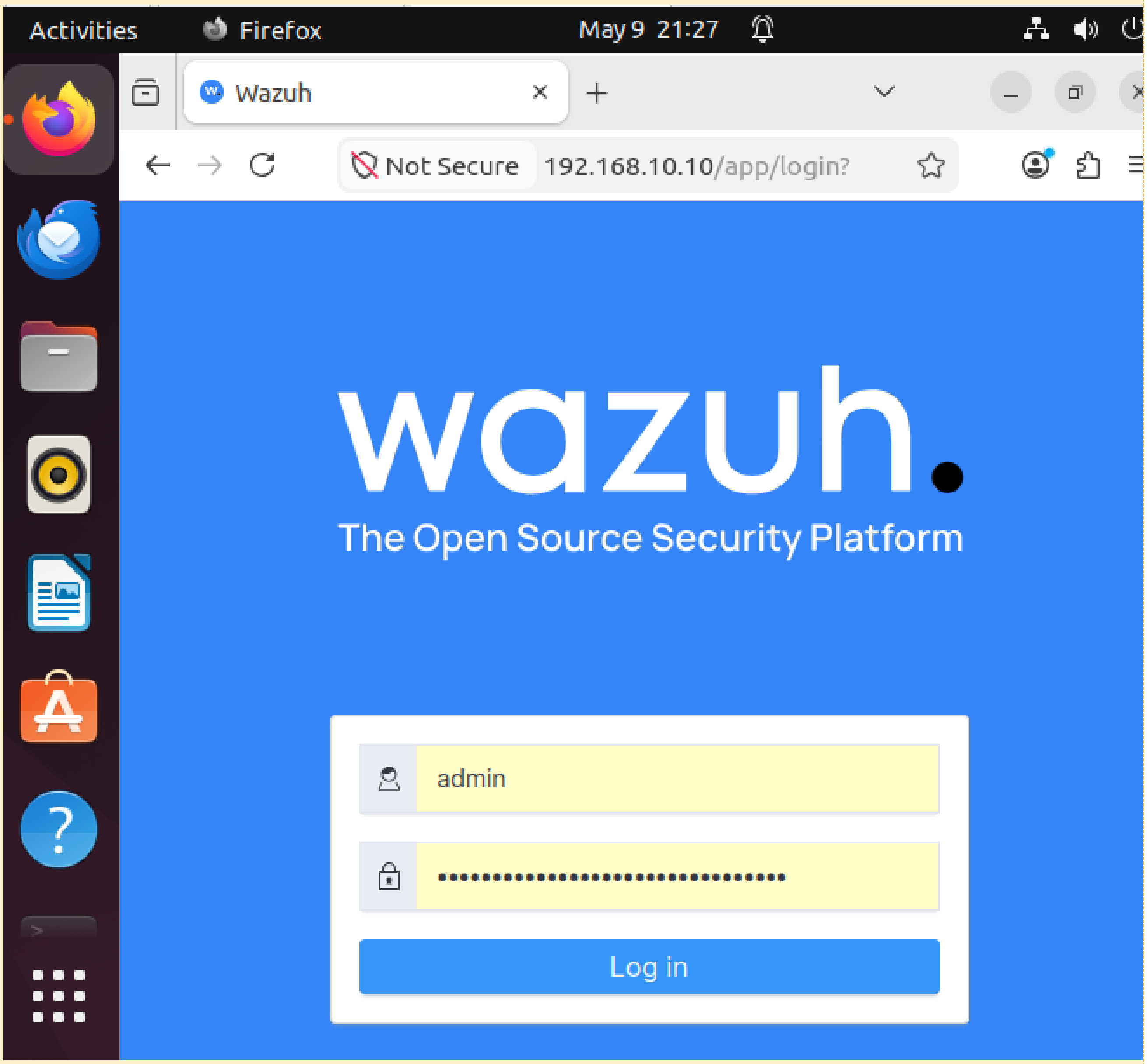
Installation Command

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash
./wazuh-install.sh -a
```

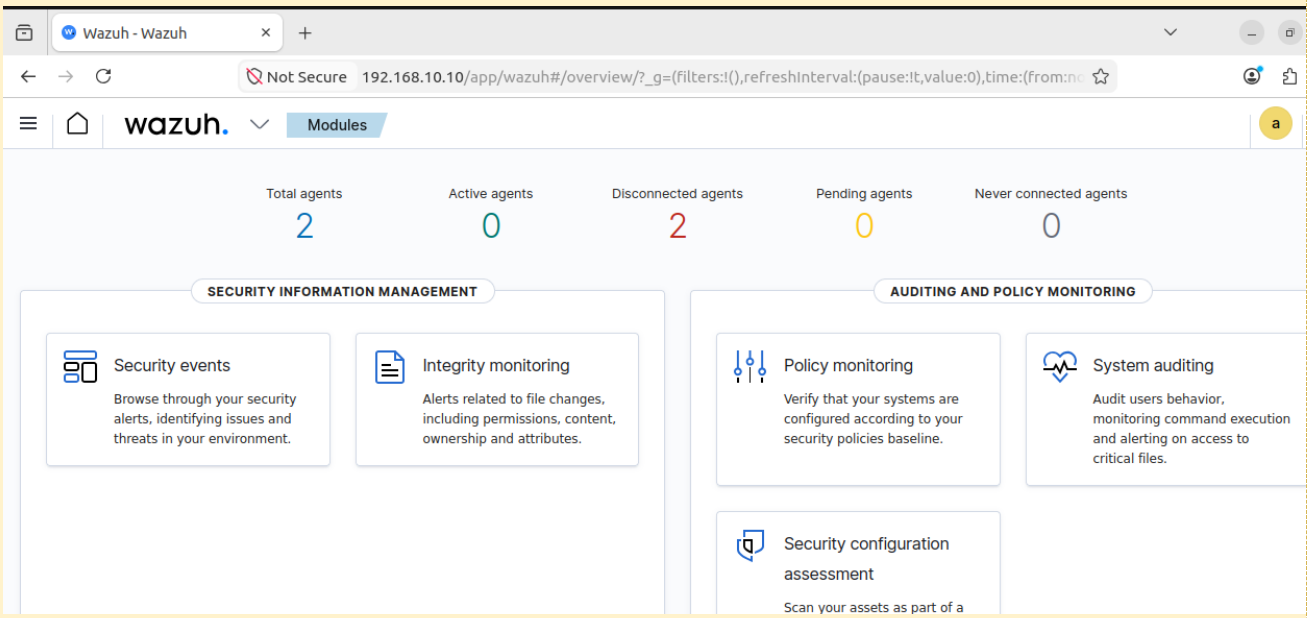
Component	Purpose	Port
Wazuh Manager	Central log collection and analysis	1514 / 1515
Wazuh Indexer	Log storage powered by OpenSearch	9200
Wazuh Dashboard	Web-based SIEM interface	443
Wazuh API	RESTful management API	55000



```
29/10/2023 14:45:06 INFO: Starting Wazuh installation assistant. Wazuh version: 4.5.4
29/10/2023 14:45:06 INFO: Verbose logging redirected to /var/log/wazuh-install.log
29/10/2023 14:45:17 INFO: --- Dependencies ----
29/10/2023 14:45:17 INFO: Installing gawk.
29/10/2023 14:45:24 INFO: Wazuh web interface port will be 443.
29/10/2023 14:45:31 INFO: --- Dependencies ----
29/10/2023 14:45:31 INFO: Installing apt-transport-https.
29/10/2023 14:45:43 INFO: Wazuh repository added.
29/10/2023 14:45:43 INFO: --- Configuration files ---
29/10/2023 14:45:43 INFO: Generating configuration files.
29/10/2023 14:45:44 INFO: Created wazuh-install-files.tar. It contains the Wazuh clus
```



Screenshot: Wazuh Dashboard login page at https://192.168.10.10



Screenshot: Wazuh Dashboard home screen after successful login

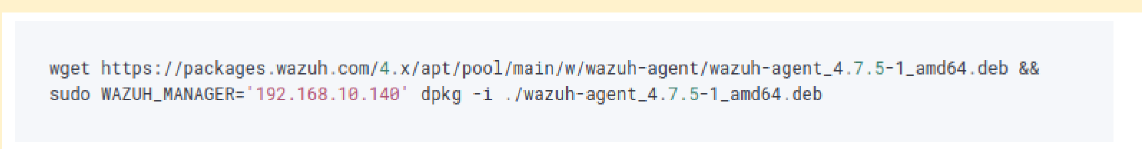
3.2 Wazuh Agent Deployment

Wazuh agents were installed on all endpoint machines to collect and forward logs to the Wazuh Manager (192.168.10.10).

Agent Installation on iRedMail Server (192.168.10.140)

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
&& sudo WAZUH_MANAGER='192.168.10.10' WAZUH_AGENT_NAME='NECMail' dpkg -i ./wazuh-agent_4.7.5-
1_amd64.deb
```

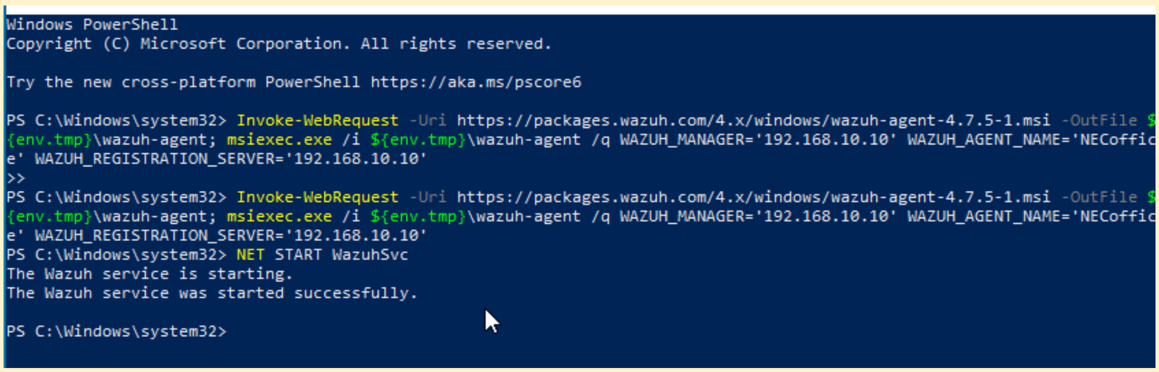
- sudo systemctl daemon-reload
- sudo systemctl enable wazuh-agent
- sudo systemctl start wazuh-agent



Screenshot: Wazuh agent installation on iRedMail server — terminal output

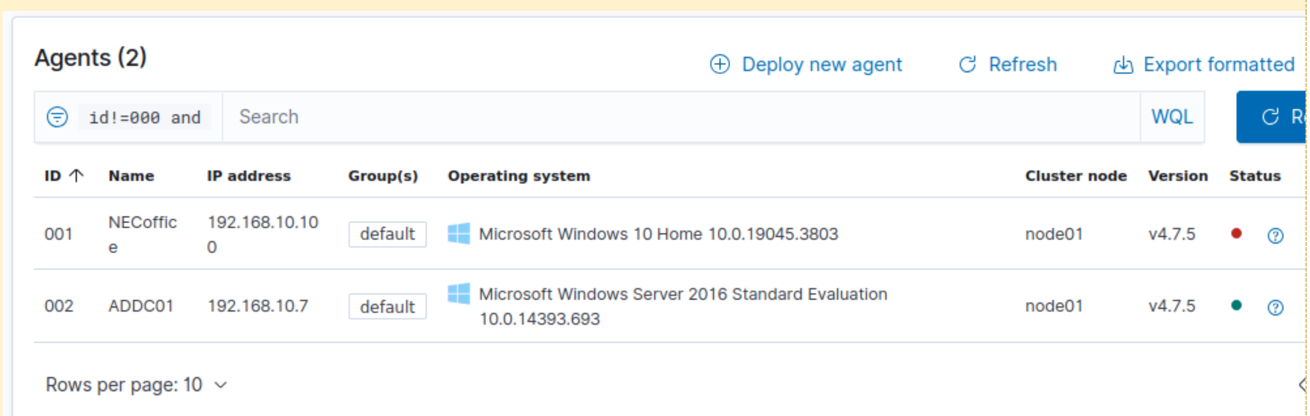
Agent Installation on Windows Server 2016 and Windows 10 Client

- Downloaded Wazuh agent MSI installer from https://packages.wazuh.com
- Set WAZUH_MANAGER environment variable to 192.168.10.10
- Completed installation wizard
- Verified agent service running in Windows Services



Screenshot: Wazuh agent MSI installer running on Windows Server 2016

3.3 Verifying Agents in Wazuh Dashboard



Screenshot: Wazuh Dashboard — Agents page showing all connected agents (iRedMail, DC01, WIN10-CLIENT)

Agent Name	IP Address	OS	Status
NECMail	192.168.10.140	Ubuntu 22.04	Active
DC01	192.168.10.7	Windows Server 2016	Active
WIN10-CLIENT	192.168.10.100	Windows 10	Active

Agent Name	IP Address	OS	Status
ubuntu-client	192.168.10.22	Ubuntu 22.04	Active

3.4 Log Sources Integrated

Log Source	Log File / Event Source	Collection Method
iRedMail — Postfix	/var/log/mail.log	Wazuh Agent
iRedMail — Dovecot	/var/log/dovecot.log	Wazuh Agent
iRedMail — Fail2Ban	/var/log/fail2ban.log	Wazuh Agent
iRedMail — Nginx	/var/log/nginx/access.log	Wazuh Agent
Windows Server 2016	Windows Security Event Log	Wazuh Agent
Windows Server 2016 — FTP	IIS FTP Logs	Wazuh Agent
Windows 10 Client	Windows Security Event Log	Wazuh Agent
Linux Client	/var/log/auth.log	Wazuh Agent

4. Demonstration of Malicious Activity Detection

To validate the SIEM's detection capabilities, the following attack scenarios were simulated from the Kali Linux attacker machine.

4.1 Brute-Force Attack on Active Directory

Detail	Description
Attack Type	Brute-force password attack against domain user accounts
Attacker	Kali Linux (192.168.10.x)
Target	Windows Server 2016 Domain Controller (192.168.10.7)
Tool Used	Hydra / Metasploit
Protocol Targeted	SMB / RDP / LDAP
Wazuh Rule Triggered	Multiple failed authentication attempts (Event ID 4625)
Alert Level	High
Detection Time	Real-time

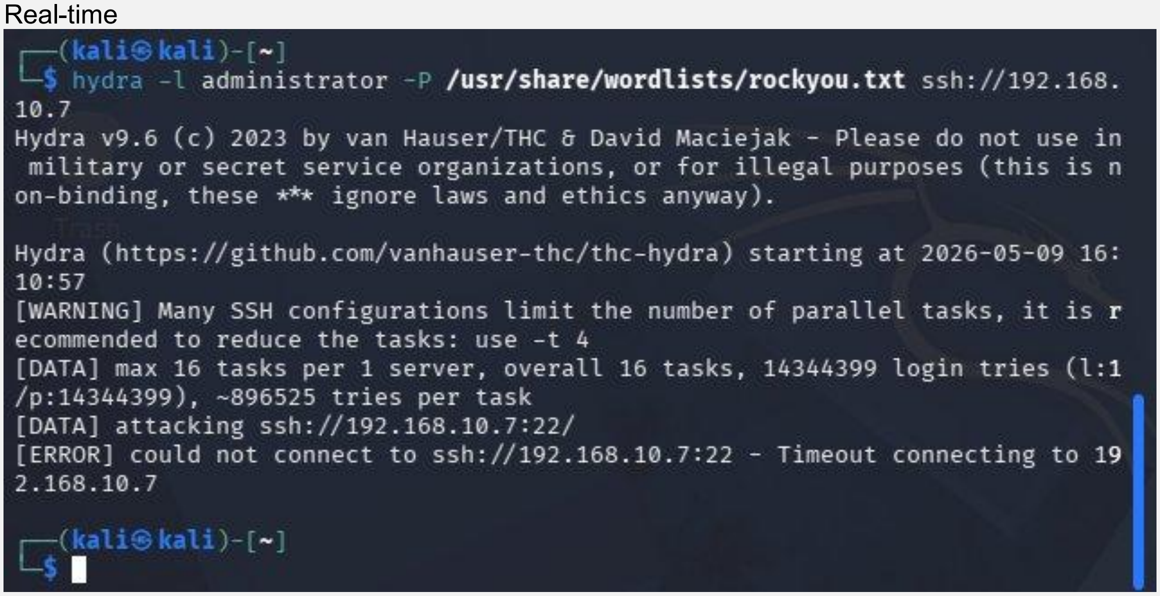
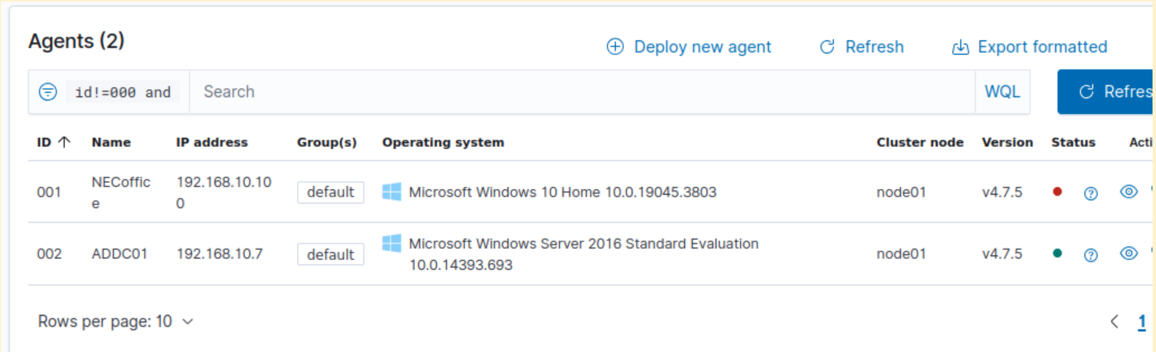


Figure: Hydra brute-force SSH attack against 192.168.10.7 from Kali Linux



Screenshot: Wazuh Dashboard — Alert showing multiple failed logins from attacker IP

4.2 Unauthorized FTP Access Attempts

Detail	Description
Attack Type	Unauthorized FTP login attempts with invalid credentials
Attacker	Kali Linux (192.168.10.x)
Target	FTP Server on Windows Server 2016 (192.168.10.7:21)
Tool Used	Hydra FTP module
Wazuh Rule Triggered	FTP authentication failure detected
Alert Level	Medium
Detection Time	Real-time

4.3 Suspicious Email with Malicious Attachment

Detail	Description
Attack Type	Phishing email with malware/suspicious attachment
Attacker	External or internal simulated sender
Target	NEC employee email accounts (john.smith@nec.local)
Detection Method	ClamAV antivirus + Amavis + Wazuh mail log monitoring
Wazuh Rule Triggered	Suspicious email attachment detected
Alert Level	High
Detection Time	Real-time

5. Response and Remediation

5.1 Incident Response Workflow

Step	Action	Tool
1	Alert generated in Wazuh Dashboard	Wazuh Dashboard
2	Security analyst reviews alert details and severity	Wazuh Dashboard
3	Investigate event logs and attack timeline	Wazuh Log Viewer
4	Identify attacker IP address and targeted system	Wazuh Dashboard
5	Block malicious IP address	Windows Firewall / UFW
6	Isolate compromised endpoint if necessary	VMware Network Settings
7	Reset compromised credentials	Active Directory
8	Implement additional access controls	AD Group Policy / FTP Settings
9	Document incident and actions taken	Incident Report

5.2 Response Actions Taken

Brute-Force Attack Response

- Identified attacker IP from Wazuh alert details
- Blocked attacker IP using Windows Firewall rule on DC01
- Enabled Account Lockout Policy after 5 failed login attempts
- Reset affected domain user passwords
- Reviewed Active Directory security logs for further compromise

FTP Unauthorized Access Response

- Identified unauthorized FTP access attempts from Wazuh alerts
- Blocked attacker IP via Windows Firewall
- Reviewed and tightened FTP directory permissions per department
- Disabled anonymous FTP access

Phishing Email Response

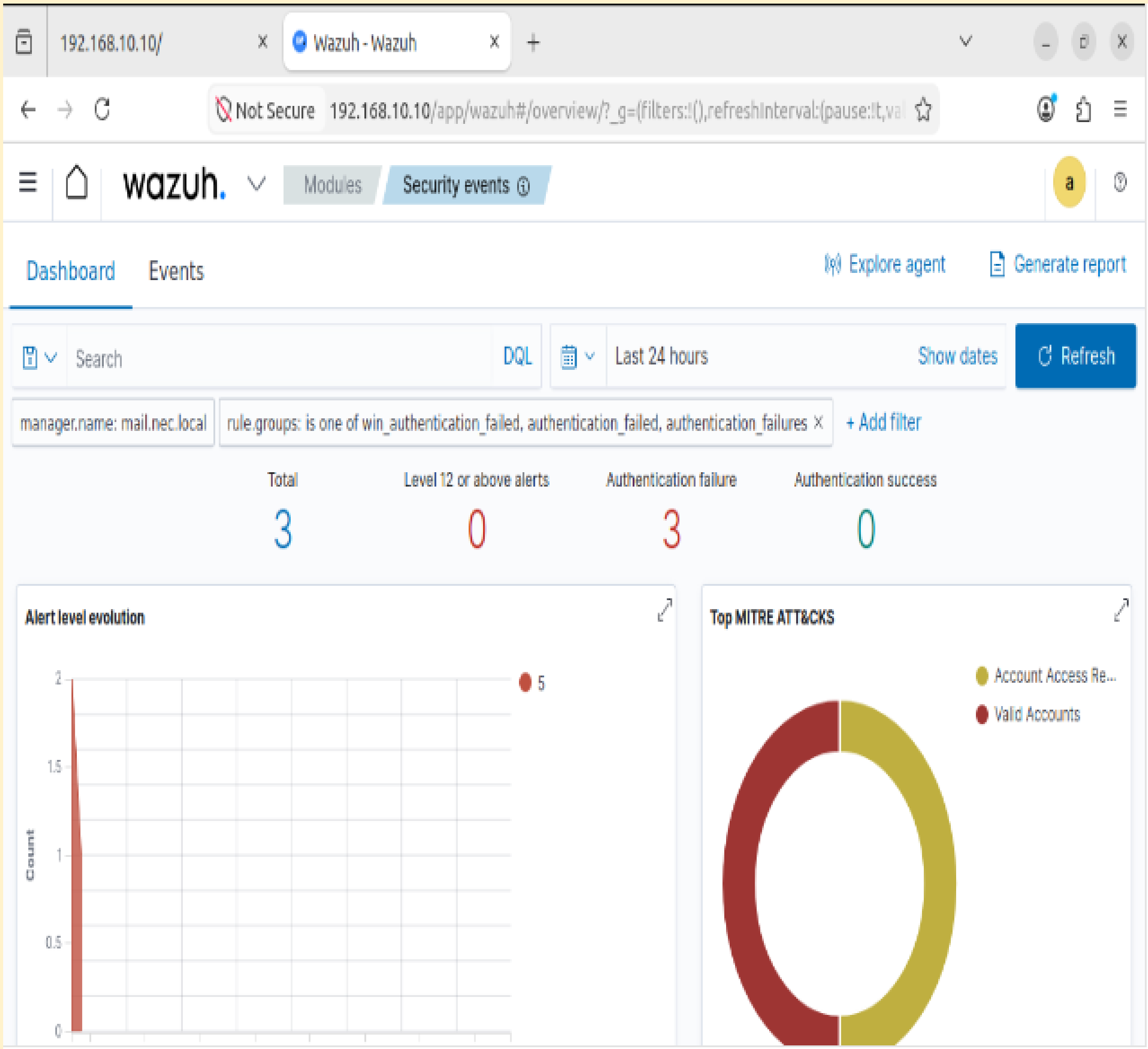
- ClamAV detected and quarantined the malicious email attachment
- Amavis logged the blocked email in /var/log/mail.log
- Added sender address to iRedMail blacklist via iRedAdmin

- Notified affected users about the phishing attempt

6. Documentation and Reporting

6.1 SIEM Effectiveness Summary

Capability	Status	Evidence
Centralised Log Collection	Implemented	All agents forwarding to Wazuh Manager
Real-time Brute-Force Detection	Implemented	Event ID 4625 alerts triggered
FTP Attack Detection	Implemented	IIS logs monitored by Wazuh
Email Threat Detection	Implemented	Mail logs + ClamAV integrated
Automated Alerting	Implemented	Wazuh dashboard alerts generated
Incident Response	Demonstrated	IP blocking and lockout policy applied
Compliance Reporting	Available	Wazuh built-in compliance dashboards (PCI DSS, GDPR)



Screenshot: Wazuh Dashboard — Security Events overview showing detected threats

6.2 Security Recommendations for NEC

Priority	Recommendation	Rationale
Critical	Implement Multi-Factor Authentication (MFA)	Prevents account compromise even if passwords are stolen
Critical	Enable TLS encryption for all email	Protects corporate communications from interception

Priority	Recommendation	Rationale
High	Implement regular patch management	Closes known vulnerabilities across all systems
High	Network segmentation (VLANs)	Isolates critical servers from general client traffic
High	Deploy Intrusion Detection System (IDS)	Additional layer of threat detection alongside SIEM
Medium	Security awareness training for staff	Reduces risk of phishing and social engineering attacks
Medium	Regular backup and disaster recovery testing	Ensures business continuity after security incidents
Low	Implement Data Loss Prevention (DLP)	Prevents sensitive data from leaving the NEC network

6.3 Wazuh Production Deployment Guidance

The following guidance is provided to NEC's IT team for full production deployment of the Wazuh SIEM:

- Allocate dedicated hardware: minimum 8GB RAM, 4 CPU cores, 500GB SSD storage for Wazuh server
- Configure log retention policy: minimum 90 days, 1 year for compliance requirements
- Set up email notifications for critical and high severity Wazuh alerts
- Regularly update Wazuh rules, decoders, and vulnerability database
- Integrate Wazuh with NEC's IT ticketing system for automated incident tracking
- Train IT security staff on Wazuh dashboard navigation and alert investigation
- Schedule quarterly security reviews using Wazuh compliance reports
- Implement Wazuh active response for automated blocking of known malicious IPs

7. Conclusion

This Proof of Concept has successfully demonstrated the effectiveness and value of the Wazuh SIEM solution in addressing NEC's cybersecurity needs. The POC validated the SIEM's capability to:

- Collect and centralise security logs from all network devices including the Domain Controller, email server, FTP server, and client machines
- Detect malicious activities in real time including brute-force attacks against Active Directory, unauthorised FTP access attempts, and phishing emails with suspicious attachments
- Provide security analysts with actionable alerts, detailed event logs, and attack timelines for efficient investigation
- Support a structured incident response workflow from alert generation through to remediation actions
- Generate compliance and security posture reports through Wazuh's built-in dashboards

The implementation demonstrates that the proposed SIEM solution will significantly improve NEC's ability to detect, respond to, and recover from cybersecurity incidents. It is strongly recommended that NEC proceeds with the full production deployment of Wazuh, following the recommendations and deployment guidance outlined in this report, to ensure the ongoing protection of NEC's network infrastructure, customer data, and business operations.